

A Distribution-Theoretical Approach to Certain Lebesgue and Sobolev Spaces*

PETER WERNER

Mathematisches Institut A, Universität Stuttgart, W.-Germany

Submitted by C. L. Dolph

The main object of this paper is to discuss some basic operations in the Hilbert space L_2 and in related Sobolev spaces by using only concepts of distribution theory without referring to Lebesgue integration. Special emphasis will be put on those aspects of the theory which seem to be relevant for applications to partial differential equations and spectral analysis of differential operators.

CONTENTS

0. Notational Conventions	18
1. Introduction	19
2. L_2 -Functionals	22
3. Derivatives and Primitives	28
4. Sobolev Spaces	35
5. Products with Continuous Functions	38
6. Fourier Transforms	42
7. Fourier Series Expansions	46
8. Translations	49
9. Mollifiers	56
10. Approximation by C^∞ -Functions	61
11. Imbedding Theorems	67
12. Change of Variables	72

0. NOTATIONAL CONVENTIONS

Let $R_n = \{x = (x_1, \dots, x_n) : x_i \text{ real}\}$ be the Euclidean n -space, topologized as usual by the inner product $x \cdot y = x_1 y_1 + \dots + x_n y_n$ and the related norm $|x| = (x_1^2 + \dots + x_n^2)^{1/2}$. For every $A \subseteq R_n$ we denote the interior, the closure, and the boundary of A by $\text{int } A$, $\bar{A}$, and ∂A , respectively. $C(A)$

* This work was sponsored by the Mathematics Research Center, United States Army, Madison, Wisconsin, under Contract No. DA-31-124-ARO-D-462.

denotes the set of all complex-valued continuous functions defined on A . $C(A)$ is a linear space if the linear operations are defined by

$$(u + v)(x) = u(x) + v(x) \quad \text{and} \quad (\alpha u)(x) = \alpha u(x).$$

We shall consider the following linear subspaces of $C(A)$: (i) the set $C_b(A)$ of all bounded $u \in C(A)$, (ii) the set $C_0(A)$ of all $u \in C(R_n)$ such that the support of u , $\text{supp } u = \{x \in R_n : u(x) \neq 0\}$, is bounded and contained in A , (iii) the set $C^m(A)$ ($m = 0, 1, \dots, \infty$) of all $u \in C(A)$ such that every derivative $(\partial/\partial x_1)^{p_1} \cdots (\partial/\partial x_n)^{p_n} u$ of order $p_1 + \cdots + p_n \leq m$ exists and is continuous in $\text{int } A$, and can be continuously extended to A , (iv) the set $C_b^m(A)$ of all $u \in C^m(A)$ such that every derivative of order $p_1 + \cdots + p_n \leq m$ is bounded in $\text{int } A$, and (v) the set $C_0^m(A) = C_0(A) \cap C^m(A)$.

Throughout this paper $\Omega, \Omega', \Omega_k, \dots$ denote nonvoid open subsets of R_n . We use the inclusion $\Omega \subset \subset \Omega'$ if Ω and Ω' are open, Ω is bounded and $\bar{\Omega} \subset \Omega'$. Note that $\Omega \subset \subset \Omega'$ implies that

$$\text{dist}(\Omega, \partial\Omega') = \inf\{|x - x'|, x \in \Omega, x' \in \Omega'\} > 0.$$

All integrations are understood in the Riemann sense and performed over the whole n -space R_n if the domain of integration is not specified otherwise.

1. INTRODUCTION

Consider a nonvoid open set $\Omega \subseteq R_n$ and let $C_0(\Omega)$ be defined as above. We introduce the inner product

$$(f, g) = \int f \bar{g} \, dx \quad (1.1)$$

for $f, g \in C_0(\Omega)$ and the related norm

$$\|f\| = (f, f)^{1/2} = \left[\int |f|^2 \, dx \right]^{1/2}. \quad (1.2)$$

As simple examples show, $C_0(\Omega)$ is not complete with respect to the norm (1.2).

In order to apply methods of abstract Hilbert space theory to problems in analysis, it is desirable to extend the inner product space $C_0(\Omega)$ to a suitable Hilbert space. Such an extension can be obtained in several ways, for example, by using Cantor's method of Cauchy sequences. However, for the purposes of analysis it is essential to choose the completion of $C_0(\Omega)$ so that the main operations of classical analysis such as integration, differentiation, and Fourier transform can be extended to the corresponding complete space. The classical way of providing such an extension of $C_0(\Omega)$ consists in introducing the Hilbert space $L_2(\Omega)$ of all equivalence classes of square-

integrable functions in the sense of Lebesgue where two functions are defined to be equivalent whenever they differ only on a set of Lebesgue measure zero. By using Lebesgue's theory of integration, the basic operations of analysis can be performed in this space. The combination of Hilbert space methods with techniques of classical analysis has been shown to be very useful. As two major applications we mention the spectral theory for ordinary and partial differential operators and the method of orthogonal projection (also known as L_2 -method) for solving boundary value problems for partial differential equations. Most of the extensive literature on these subjects makes relevant use of Lebesgue's integration theory.

In several recent papers a strong tendency is visible to reduce the references to classical Lebesgue theory by employing a more abstract point of view. As an example, we refer to lecture notes of P. D. Lax on the theory of functions of a real variable [3] where the space L_p is introduced as the Cauchy completion of the space of continuous functions with respect to the p -norm. Lax points out that several properties of the space L_p can be discussed on this basis without using Lebesgue integration and introduces the concept of the Lebesgue integral at a later stage.

In this paper we discuss a similar approach to the theory of the Hilbert space $L_2(\Omega)$ which is based on elementary concepts of L. Schwartz's theory of distributions [6] and does not require Lebesgue's theory of measure and integration. Let us recall that a distribution in Ω is defined as a linear functional on $C_0^\infty(\Omega)$ satisfying a certain continuity property. Our starting point is the remark that every Lebesgue-integrable function u defines a distribution F_u in the sense of L. Schwartz by

$$F_u \varphi = \int u \varphi \, dx \quad \text{for} \quad \varphi \in C_0^\infty(\Omega) \quad (1.3)$$

where the integral is interpreted in Lebesgue's sense. Two Lebesgue-integrable functions define the same linear functional by (1.3) if and only if they differ only on a set of Lebesgue measure zero. This implies that (1.3) yields a one-to-one linear mapping of the space $L_2(\Omega)$ of all equivalence classes of square-integrable functions into the linear space $D'(\Omega)$ of all distributions in Ω . By identifying the equivalence class $\bar{u}$ of u with the functional F_u defined by (1.3), $L_2(\Omega)$ can be considered as a subspace of $D'(\Omega)$. As shown by L. Schwartz [6], the main operations of classical calculus can be extended to $D'(\Omega)$ in a very natural and elementary way. This suggests that it ought to be feasible to define and to discuss the basic operations in the Hilbert space $L_2(\Omega)$ by using the corresponding definitions of distribution theory. For this point of view which will be adopted throughout this paper it is essential that those distributions which correspond to L_2 -functions by (1.3) can be characterized without using Lebesgue integration. For example,

it is easy to show that the distributions generated by L_2 -functions are identical with those linear functionals on $C_0^\infty(\Omega)$ which are bounded with respect to the norm (1.2). This observation furnishes the basis of the approach used in this paper.

In the next section we shall introduce $L_2(\Omega)$ as the conjugate space of the linear space $C_0^\infty(\Omega)$ normed by (1.2). It follows immediately that $L_2(\Omega)$ is complete with respect to the norm

$$\|F\| = \text{Sup}\{ |F\varphi| : \varphi \in C_0^\infty(\Omega), \|\varphi\| = 1 \}. \quad (1.4)$$

A simple application of the representation theorem of Riesz for bounded linear functionals in a Hilbert space shows that the norm (1.4) can be generated by an inner product and that the set of linear functionals defined for functions $u \in C_0^\infty(\Omega)$ by means of formula (1.3) is dense in $L_2(\Omega)$. This verifies that the space $L_2(\Omega)$ defined in Section 2 is isometric to the classical space $L_2(\Omega)$.

In the remaining sections we shall discuss the main operations of the L_2 -calculus such as differentiation, integration, multiplication, Fourier series expansion, Fourier transform, translation, change of variables, approximation by smooth functions, and imbedding of Sobolev spaces, by using definitions suggested by distribution theory. We intend to carry the analysis to a point which allows us to discuss the standard applications to partial differential equations and spectral theory. In subsequent papers we plan to discuss some of these applications by using the concepts developed in this paper.

We emphasize that most of the material covered in this paper is well-known in the usual framework of Lebesgue's theory. Our principal aim is not to add new results to the classical theory of Lebesgue and Sobolev spaces, but to introduce a new method of presentation which replaces the use of Lebesgue's measure and integration theory completely by functional-analytical and distribution-theoretical considerations. We have the feeling that the approach discussed here adds to the transparency of the theory and simplifies several applications to problems in partial differential equations and spectral theory. In view of future references we found it convenient to give a self-contained representation which uses only elementary (pre-Lebesgue) calculus and standard theorems of Hilbert space theory. In particular, we have also included those parts of the proofs which are related to familiar arguments in the classical approach. We refer to the first chapters of [1] and [8] for the theory of Sobolev spaces on the basis of Lebesgue's integration theory. A brief introduction to the terminology and the principal results of distribution theory is given in the first chapter of [2].

The methods of this paper can be used for a similar discussion of the theory of L_p -spaces. For example, the space $L_p(\Omega)$, $1 < p < \infty$, can be introduced as the conjugate of $C_0^\infty(\Omega)$ with respect to the q -norm where

$1/p + 1/q = 1$. However, in view of the applications to partial differential equations which usually require only the consideration of Hilbert spaces, we have restricted our analysis to the space $L_2(\Omega)$ and the related Sobolev spaces. Extensions to L_p -spaces and applications to the Fourier-transform in $L_1(-\infty, \infty)$ will be discussed in [9].

2. L_2 -FUNCTIONALS

As pointed out in the introduction, we introduce the space $L_2(\Omega)$ as the conjugate of the linear space $C_0^\infty(\Omega)$ normed by (1.2):

DEFINITION 2.1. $L_2(\Omega)$ is the linear space of all complex-valued linear functionals F on $C_0^\infty(\Omega)$ such that

$$\|F\| = \sup\{|F\varphi| : \varphi \in C_0^\infty(\Omega), \|\varphi\| = 1\} < \infty, \quad (2.1)$$

with the linear operations defined as usual by

$$(F_1 + F_2)\varphi = F_1\varphi + F_2\varphi, \quad (cF)\varphi = cF\varphi. \quad (2.2)$$

Every element of $L_2(\Omega)$ is called a L_2 -functional.

$L_2(\Omega)$ is a normed linear space with the norm (2.1). As a conjugate of a normed linear space, $L_2(\Omega)$ is complete. In fact, if $\{F_k\}$ is a sequence of L_2 -functionals with $\|F_j - F_k\| \rightarrow 0$ as $j, k \rightarrow \infty$, then the functional F defined by

$$F\varphi = \lim_{k \rightarrow \infty} F_k\varphi \quad (2.3)$$

belongs also to $L_2(\Omega)$ and satisfies $\|F_j - F\| \rightarrow 0$ as $j \rightarrow \infty$.

We show that $C_0^\infty(\Omega)$ can be considered as a subspace of $L_2(\Omega)$. For every $\psi \in C_0^\infty(\Omega)$ consider the linear functional F_ψ defined by

$$F_\psi\varphi = \int \varphi\psi \, dx \quad \text{for} \quad \varphi \in C_0^\infty(\Omega). \quad (2.4)$$

By Schwarz's inequality, we obtain $|F_\psi\varphi| \leq \|\varphi\| \cdot \|\psi\|$ and hence $F_\psi \in L_2(\Omega)$ and $\|F_\psi\| \leq \|\psi\|$. By setting $\varphi = \bar{\psi}/\|\psi\|$ in (2.4), it follows that $\|F_\psi\| = \|\psi\|$. In particular, $F_\psi = 0$ implies $\psi = 0$. This shows that $\psi \rightarrow F_\psi$ is a one-to-one, linear, isometric mapping from $C_0^\infty(\Omega)$ into $L_2(\Omega)$. This remark enables us to identify every $\psi \in C_0^\infty(\Omega)$ with the linear functional F_ψ defined by (2.4). Thus, $C_0^\infty(\Omega)$ can be considered as a subspace of $L_2(\Omega)$. In the sequel we shall replace F_ψ by ψ whenever it is convenient. In particular, we shall write $F - \psi$ and $\|F - \psi\|$ for $F \in L_2(\Omega)$ and $\psi \in C_0^\infty(\Omega)$ instead of $F - F_\psi$ and $\|F - F_\psi\|$.

To show that $C_0^\infty(\Omega)$ is dense in $L_2(\Omega)$, we introduce the linear space $\overline{C_0^\infty(\Omega)}$ consisting of all L_2 -functionals F with the property that a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ exists such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. We show:

LEMMA 2.1. $\overline{C_0^\infty(\Omega)}$ is complete.

PROOF. Consider a sequence $\{F_k\}$ in $\overline{C_0^\infty(\Omega)}$ such that $\|F_j - F_k\| \rightarrow 0$ as $j, k \rightarrow \infty$. By definition of $\overline{C_0^\infty(\Omega)}$, we can choose functions $f_k \in C_0^\infty(\Omega)$ such that $\|F_k - f_k\| < 1/k$ for every k . By the triangle inequality,

$$\|f_j - f_k\| \leq \|F_j - F_k\| + \frac{1}{j} + \frac{1}{k} \rightarrow 0 \quad \text{as } j, k \rightarrow \infty. \quad (2.5)$$

Hence the sequence $\{f_k\}$ is convergent in $L_2(\Omega)$. Since $L_2(\Omega)$ is complete, there exists a $F \in L_2(\Omega)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. By definition of $\overline{C_0^\infty(\Omega)}$, F belongs to $\overline{C_0^\infty(\Omega)}$. A second application of the triangle inequality yields

$$\|F - F_k\| \leq \|F - f_k\| + \frac{1}{k} \rightarrow 0 \quad \text{as } k \rightarrow \infty, \quad (2.6)$$

which proves the completeness of $\overline{C_0^\infty(\Omega)}$.

Since $C_0^\infty(\Omega)$ is dense in $\overline{C_0^\infty(\Omega)}$, the inner product in $C_0^\infty(\Omega)$ defined by (1.1) can be extended to $\overline{C_0^\infty(\Omega)}$. In fact, assume that F and G belong to $\overline{C_0^\infty(\Omega)}$ and that $\{f_k\}$ and $\{g_k\}$ are sequences in $C_0^\infty(\Omega)$ converging to F and G , respectively. Then an inner product (F, G) can be defined by

$$(F, G) = \lim_{k \rightarrow \infty} (f_k, g_k) = \lim_{k \rightarrow \infty} \int f_k \cdot \bar{g}_k \, dx. \quad (2.7)$$

The limit on the right-hand side exists since, by Schwarz's inequality,

$$|(f_k, g_k) - (f_j, g_j)| \leq \|f_k - f_j\| \cdot \|g_k\| + \|f_j\| \cdot \|g_k - g_j\| \rightarrow 0 \quad \text{as } k, j \rightarrow \infty.$$

Assume that the sequences $\{f'_k\}$ and $\{g'_k\}$ in $C_0^\infty(\Omega)$ also converge to F and G , respectively. Then $\|f_k - f'_k\| \rightarrow 0$ and $\|g_k - g'_k\| \rightarrow 0$ as $k \rightarrow \infty$. Hence,

$$|(f_k, g_k) - (f'_k, g'_k)| \leq \|f_k - f'_k\| \cdot \|g_k\| + \|f'_k\| \cdot \|g_k - g'_k\| \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

This shows that the limit (2.7) is independent of the special choice of the sequences $\{f_k\}$ and $\{g_k\}$ converging to F and G . Thus, the product definition (2.7) is justified for $F, G \in \overline{C_0^\infty(\Omega)}$.

We show that the norm (2.1) and the product (2.7) are related by

$$\|F\| = (F, F)^{1/2} \quad \text{for } F \in \overline{C_0^\infty(\Omega)}. \quad (2.8)$$

Choose a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ with $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. The triangle inequality implies $\|f_k\| \rightarrow \|F\|$ as $k \rightarrow \infty$. On the other hand, (2.7) shows that $\|f_k\|^2 = (f_k, f_k) \rightarrow (F, F)$ as $k \rightarrow \infty$. In particular, $(F, F) = 0$ implies $F = 0$. The remaining properties required for an inner product follow from (2.7).

Relation (2.8) says that, for elements of $\overline{C_0^\infty(\Omega)}$, the norm (2.1) is generated by the inner product (2.7). Since $\overline{C_0^\infty(\Omega)}$ is complete with respect to the norm (2.1) by Lemma 2.1, it follows that $\overline{C_0^\infty(\Omega)}$ is a Hilbert space with the inner product (2.7).

After these preparations we show:

LEMMA 2.2.

$$L_2(\Omega) = \overline{C_0^\infty(\Omega)}.$$

PROOF. Assume that F belongs to $L_2(\Omega)$, i.e., that F is a bounded linear functional on the linear space $C_0^\infty(\Omega)$ normed by (1.2). Since $C_0^\infty(\Omega)$ is dense in $\overline{C_0^\infty(\Omega)}$, F can be extended to a bounded linear functional on $\overline{C_0^\infty(\Omega)}$, by setting

$$FG = \lim_{k \rightarrow \infty} Fg_k \quad (2.9)$$

where $G \in \overline{C_0^\infty(\Omega)}$, $g_k \in C_0^\infty(\Omega)$ and $\|G - g_k\| \rightarrow 0$ as $k \rightarrow \infty$. Note that, by (2.1), the limit in (2.9) exists and is independent of the special choice of the sequence $\{g_k\}$ converging to G .

Since $\overline{C_0^\infty(\Omega)}$ is a Hilbert space, the theorem of F. Riesz on the representation of bounded linear functionals implies that a functional $H \in \overline{C_0^\infty(\Omega)}$ exists such that

$$FG = (G, H) \quad \text{for every} \quad G \in \overline{C_0^\infty(\Omega)}. \quad (2.10)$$

By definition of $\overline{C_0^\infty(\Omega)}$, a sequence $\{h_k\}$ exists in $C_0^\infty(\Omega)$ such that $\|H - h_k\| \rightarrow 0$ as $k \rightarrow \infty$. By the continuity of the inner product, which follows immediately from Schwarz's inequality, and by (2.10), we obtain

$$F\varphi = (\varphi, H) = \lim_{k \rightarrow \infty} (\varphi, h_k) \quad (2.11)$$

for every $\varphi \in C_0^\infty(\Omega)$. Hence

$$\begin{aligned} \|F - h_j\| &= \sup_{\|\varphi\|=1} \left| F\varphi - \int \varphi h_j dx \right| = \sup_{\|\varphi\|=1} \left| \lim_{k \rightarrow \infty} (\varphi, h_k - h_j) \right| \\ &\leq \lim_{k \rightarrow \infty} \|h_k - h_j\| \rightarrow 0 \quad \text{as} \quad j \rightarrow \infty. \end{aligned}$$

This proves that F is the limit of the sequence $\{h_j\}$ in $C_0^\infty(\Omega)$ and hence belongs to $\overline{C_0^\infty(\Omega)}$. This concludes the proof of Lemma 2.2.

Lemma 2.2 shows that $L_2(\Omega)$ is a Hilbert space and that $C_0^\infty(\Omega)$ is a dense linear subspace of $L_2(\Omega)$. We collect the main results of our analysis in the following theorem:

THEOREM 2.1. *By identifying the function φ of $C_0^\infty(\Omega)$ with the functional F_φ defined by (2.4), the linear space $C_0^\infty(\Omega)$ normed by (1.2) is isometrically*

imbedded into the linear space $L_2(\Omega)$ introduced by Definition 2.1 and normed by (2.1). $C_0^\infty(\Omega)$ is a dense subspace of $L_2(\Omega)$ with respect to the norm (2.1). An inner product can be introduced in $L_2(\Omega)$ by formula (2.7) where $f_k, g_k \in C_0^\infty(\Omega)$ and $\|F - f_k\| \rightarrow 0, \|G - g_k\| \rightarrow 0$ as $k \rightarrow \infty$. $L_2(\Omega)$ is a Hilbert space with respect to this inner product. The norm (2.1) and the inner product (2.7) are related by formula (2.8).

Theorem 2.1 says that $L_2(\Omega)$ is the completion of the inner product space $C_0^\infty(\Omega)$. This implies, by familiar results of Lebesgue theory, that $L_2(\Omega)$ is isometric to the classical space of equivalence classes of square-integrable functions in the sense of Lebesgue.

As a simple corollary of Theorem 2.1, we obtain:

LEMMA 2.3. For every $F \in L_2(\Omega)$ and every $\varphi \in C_0^\infty(\Omega)$, the relation

$$F\varphi = (F, \bar{\varphi}) \quad (2.12)$$

holds.

In fact, by Theorem 2.1 a sequence $\{f_k\}$ exists in $C_0^\infty(\Omega)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. Hence we obtain, by (1.1) and the continuity of the inner product,

$$F\varphi = \lim_{k \rightarrow \infty} \int f_k \varphi \, dx = \lim_{k \rightarrow \infty} (f_k, \bar{\varphi}) = (F, \bar{\varphi}).$$

As the next step of our analysis, we discuss the imbedding of continuous functions in $L_2(\Omega)$. We say that a function $u \in C(\Omega)$ belongs to $L_2(\Omega)$ if the functional F_u defined by (1.3) belongs to $L_2(\Omega)$. If $u \in C(\Omega) \cap L_2(\Omega)$ in this sense, then the values $u(x)$ of u at all points $x \in \Omega$ are uniquely determined by the functional F_u as the following Lemma shows:

LEMMA 2.4. Assume that $u \in C(\Omega)$ and that

$$F_u \varphi = \int u \varphi \, dx = 0 \quad \text{for every} \quad \varphi \in C_0^\infty(\Omega).$$

Then $u(x) = 0$ for every $x \in \Omega$.

PROOF. Choose a function h such that

$$h \in C_0^\infty(R_n), \quad \text{supp } h = \{x : |x| \leq 1\}, \quad h \geq 0, \quad \int h \, dx = 1. \quad (2.13)$$

For example, the function

$$h(x) = \begin{cases} ce^{1/(|x|^2-1)} & \text{for } |x| < 1, \\ 0 & \text{for } |x| \geq 1 \end{cases} \quad (2.14)$$

has, for suitable c , the required properties. Consider a point $x_0 \in \Omega$. Since Ω is open, we can choose a $\delta_0 > 0$ such that the ball $\{x : |x - x_0| \leq \delta_0\}$ is contained in Ω . If $\delta < \delta_0$, the function

$$h_\delta(x) = \frac{1}{\delta^n} h\left(\frac{x - x_0}{\delta}\right) \quad (2.15)$$

belongs to $C_0^\infty(\Omega)$. By setting $x = x_0 + \delta y$, we obtain

$$\begin{aligned} F_u h_\delta &= \frac{1}{\delta^n} \int u(x) h\left(\frac{x - x_0}{\delta}\right) dx = \int_{|y| \leq 1} u(x_0 + \delta y) h(y) dy \\ &\rightarrow u(x_0) \int h(y) dy = u(x_0) \quad \text{as } \delta \rightarrow 0 \end{aligned}$$

and hence $u(x_0) = 0$ since $F_u h_\delta = 0$ for $0 < \delta < \delta_0$. This concludes the proof of Lemma 2.4.

Lemma 2.4 enables us to identify each function $u \in C(\Omega) \cap L_2(\Omega)$ with the corresponding functional F_u . In particular, we set $\|u\| = \|F_u\|$ and $(u, v) = (F_u, F_v)$ if $u, v \in C(\Omega) \cap L_2(\Omega)$. As an example we evaluate (u, v) for $u, v \in C_0(\Omega)$. For this purpose we choose sequences $\{u_k\}, \{v_k\}$ in $C_0^\infty(\Omega)$ such that

$$\int |u - u_k|^2 dx \rightarrow 0, \quad \int |v - v_k|^2 dx \rightarrow 0 \quad \text{as } k \rightarrow \infty. \quad (2.16)$$

A sequence $\{u_k\}$ with this property can be obtained in the following way. Set

$$u_k(x) = k^n \int u(y) h(k(y - x)) dy = \int u\left(x + \frac{y}{k}\right) h(y) dy \quad (2.17)$$

where h is a function with the properties (2.13). Since $u \in C_0(\Omega)$, we have $\delta = \text{dist}(\text{supp } u, \partial\Omega) > 0$. Clearly u_k belongs to $C_0^\infty(\Omega)$ if $k > 1/\delta$. Since

$$u_k(x) - u(x) = \int \left[u\left(x + \frac{y}{k}\right) - u(x) \right] h(y) dy, \quad (2.18)$$

$\{u_k\}$ converges uniformly to u as $k \rightarrow \infty$. This implies the first limit relation in (2.16) since $\text{supp } u$ and $\text{supp } u_k$ are contained in the bounded set $\{x : \text{dist}(x, \text{supp } u) \leq 1\}$.

By (2.16) and Schwarz's inequality we obtain for every $\varphi \in C_0^\infty(\Omega)$ with $\|\varphi\| = 1$

$$\begin{aligned} \left| F_u \varphi - \int u_k \varphi dx \right| &= \left| \int (u - u_k) \varphi dx \right| \leq \left[\int |u - u_k|^2 dx \right]^{1/2} \rightarrow 0 \\ &\quad \text{as } k \rightarrow \infty \end{aligned}$$

and hence $\|F_u - u_k\| \rightarrow 0$ as $k \rightarrow \infty$. In the same way we conclude that $\|F_v - v_k\| \rightarrow 0$ as $k \rightarrow \infty$. This implies, by the definition of the inner product in (2.7) and Schwarz's inequality, that

$$(u, v) = (F_u, F_v) = \lim_{k \rightarrow \infty} (u_k, v_k) = \lim_{k \rightarrow \infty} \int u_k \bar{v}_k dx = \int u \bar{v} dx. \quad (2.19)$$

In particular, we obtain for $u \in C_0(\Omega)$

$$\|u\| = (u, u)^{1/2} = \left[\int |u|^2 dx \right]^{1/2}. \quad (2.20)$$

Thus we have proved:

LEMMA 2.5. *$C_0(\Omega)$ is a linear subspace of $C(\Omega) \cap L_2(\Omega)$. Furthermore, the inner product of two functions $u, v \in C_0(\Omega)$ is given by*

$$(u, v) = (F_u, F_v) = \int u \bar{v} dx. \quad (2.21)$$

A similar result holds for all functions u, v which can be approximated by sequences $\{u_k\}$ and $\{v_k\}$ in $C_0^\infty(\Omega)$ in the sense of (2.16). In particular, it is possible to replace $C_0(\Omega)$ by $C(\bar{\Omega})$ if Ω is bounded and the boundary of Ω is sufficiently smooth so as to permit Riemann integration over Ω .

It is easy to verify that every $F \in L_2(\Omega)$ is a distribution in Ω in the sense of L. Schwartz. Consider a sequence $\{\varphi_k\}$ in $C_0^\infty(\Omega)$ such that (i) the supports of all functions φ_k are contained in a fixed compact subset of Ω , (ii) φ_k and all derivatives of φ_k converge uniformly to zero as $k \rightarrow \infty$. A linear functional F on $C_0^\infty(\Omega)$ is a distribution if $F\varphi_k \rightarrow 0$ as $k \rightarrow \infty$ for every sequence $\{\varphi_k\}$ satisfying (i) and (ii). Properties (i) and (ii) imply that $\|\varphi_k\| \rightarrow 0$ as $k \rightarrow \infty$. Since $|F\varphi_k| \leq \|F\| \cdot \|\varphi_k\|$ for every $F \in L_2(\Omega)$, we conclude that $F\varphi_k \rightarrow 0$ as $k \rightarrow \infty$. Hence every functional $F \in L_2(\Omega)$ is a distribution. In particular, all operations which are considered in the theory of distributions can be specialized to L_2 -functionals. This remark motivates the analysis carried out in the following sections.

We conclude this section with some remarks on the relations between the spaces $L_2(\Omega)$ and $L_2(\Omega')$ where Ω and Ω' are two open subsets of R_n with $\Omega \subset \Omega'$. If $F \in L_2(\Omega')$, then the *restriction* F^r of the functional F to $C_0^\infty(\Omega)$ defined by

$$F^r \varphi = F\varphi \quad \text{for} \quad \varphi \in C_0^\infty(\Omega) \quad (2.22)$$

belongs to $L_2(\Omega)$, and (2.1) implies that

$$\|F^r\|_{L_2(\Omega)} \leq \|F\|_{L_2(\Omega')}. \quad (2.23)$$

On the other hand, every functional $F \in L_2(\Omega)$ can be extended to a functional $F^* \in L_2(\Omega')$ by the following procedure: Choose a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$ and set

$$F^*\varphi = \lim_{k \rightarrow \infty} \int f_k \varphi \, dx \quad \text{for} \quad \varphi \in C_0^\infty(\Omega'). \quad (2.24)$$

By Schwarz's inequality, the limit (2.24) exists and is independent of the special sequence $\{f_k\}$ used for the approximation of F . Furthermore, we have for $\varphi \in C_0^\infty(\Omega')$ and $\|\varphi\| = 1$

$$\left| F^*\varphi - \int f_j \varphi \, dx \right| \leq \lim_{k \rightarrow \infty} \|f_k - f_j\| \rightarrow 0 \quad \text{as} \quad j \rightarrow \infty$$

and hence

$$\|F^* - f_j\|_{L_2(\Omega')} \rightarrow 0 \quad \text{as} \quad j \rightarrow \infty. \quad (2.25)$$

This implies $F^* \in L_2(\Omega')$ and

$$\|F^*\|_{L_2(\Omega')} = \lim_{k \rightarrow \infty} \|f_k\| = \|F\|_{L_2(\Omega)}. \quad (2.26)$$

By (2.24) we have $F^*\varphi = F\varphi$ if $\varphi \in C_0^\infty(\Omega)$ and hence

$$(F^*)^r = F. \quad (2.27)$$

We call F^* the *norm-invariant extension* of F to $L_2(\Omega')$. The mapping $F \rightarrow F^*$ corresponds to the extension of a square-integrable function u , defined in Ω , to Ω' by setting $u(x) = 0$ if $x \in \Omega' - \Omega$.

3. DERIVATIVES AND PRIMITIVES

In this section we study derivatives of L_2 -functionals by adopting the usual definition of distribution theory. Let F be a linear functional on $C_0^\infty(\Omega)$. The derivative $(\partial/\partial x_i)F$ of F with respect to the i th coordinate direction ($i = 1, \dots, n$) is defined as the linear functional on $C_0^\infty(\Omega)$ satisfying

$$\left(\frac{\partial}{\partial x_i} F \right) \varphi = -F \left(\frac{\partial}{\partial x_i} \varphi \right) \quad \text{for} \quad \varphi \in C_0^\infty(\Omega). \quad (3.1)$$

Note that the right-hand side is defined since $(\partial/\partial x_i)\varphi$ also belongs to $C_0^\infty(\Omega)$. Hence every linear functional on $C_0^\infty(\Omega)$, and in particular every L_2 -functional, possesses derivatives in this sense. $L_2(\Omega)$ is not closed with respect to differentiation as simple examples show.

Assume that $F = F_u$ is generated as in (1.3) by a function $u \in C^1(\Omega)$. Integration by parts yields

$$\left(\frac{\partial}{\partial x_i} F_u\right) \varphi = -F_u \left(\frac{\partial \varphi}{\partial x_i}\right) = -\int u \frac{\partial}{\partial x_i} \varphi \, dx = \int \frac{\partial u}{\partial x_i} \varphi \, dx = F_{\partial u / \partial x_i} \varphi \quad (3.2)$$

and hence

$$\frac{\partial}{\partial x_i} F_u = F_{\partial u / \partial x_i}. \quad (3.3)$$

This relation shows that differentiation in the functional sense as defined above is compatible with classical differentiation.

In the remaining part of this section we discuss some properties of L_2 -functionals which are closely related to classical theorems on differentiation. It is a well-known fact that every Lebesgue-integrable function of one variable is a derivative (in the sense of Lebesgue) of a continuous function. We shall prove a similar statement by using the derivative concept introduced above. Assume that Ω is a bounded open interval of the real axis R_1 , say $\Omega = (a, b)$ with $a < b$, and consider a linear functional F belonging to $L_2(a, b)$. By Theorem 2.1, a sequence $\{f_k\}$ exists in $C_0^\infty(a, b)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. Hence

$$F\varphi = \lim_{k \rightarrow \infty} \int_a^b f_k \varphi \, dx = \lim_{k \rightarrow \infty} (f_k, \bar{\varphi}) \quad (3.4)$$

for every $\varphi \in C_0^\infty(a, b)$. A primitive U of F is given by

$$U\varphi = -\lim_{k \rightarrow \infty} \int_a^b f_k(x) \left[\int_a^x \varphi(t) \, dt \right] dx = -\lim_{k \rightarrow \infty} \left(f_k, \int_0^x \overline{\varphi(t)} \, dt \right). \quad (3.5)$$

In fact, since $\varphi(a) = 0$ for $\varphi \in C_0^\infty(a, b)$, we have

$$\left(\frac{d}{dx} U\right) \varphi = -U \left(\frac{d\varphi}{dx}\right) = \lim_{k \rightarrow \infty} \left(f_k, \int_0^x \overline{\varphi(t)} \, dt \right) = \lim_{k \rightarrow \infty} (f_k, \bar{\varphi}) = F\varphi$$

and hence

$$\frac{d}{dx} U = F. \quad (3.6)$$

To prove that U is generated by a continuous function u , we interchange the order of integrations in (3.5):

$$\int_a^b f_k(x) \left[\int_a^x \varphi(t) \, dt \right] dx = \int_a^b \varphi(t) \left[\int_t^b f_k(x) \, dx \right] dt.$$

This implies

$$U\varphi = \lim_{k \rightarrow \infty} \int_a^b \varphi(x) u_k(x) dx = \lim_{k \rightarrow \infty} (u_k, \bar{\varphi}) \quad (3.7)$$

with

$$u_k(x) = - \int_x^b f_k(t) dt. \quad (3.8)$$

By Schwarz's inequality,

$$\begin{aligned} |u_j(x) - u_k(x)| &= \left| \int_x^b [f_j(t) - f_k(t)] dt \right| \\ &\leq \sqrt{b-x} \|f_j - f_k\| \leq \sqrt{b-a} \|f_j - f_k\| \rightarrow 0 \\ &\quad \text{as } j, k \rightarrow \infty, \end{aligned} \quad (3.9)$$

which shows that $\{u_k(x)\}$ converges uniformly in the closed interval $[a, b]$. Hence, the sequence $\{u_k(x)\}$ has a continuous limit $u(x)$ in $[a, b]$ and the order of the limit and the integration in (3.7) can be interchanged yielding

$$U\varphi = \int_a^b u\varphi dx. \quad (3.10)$$

This proves that the primitive U of F is generated by the function $u \in C[a, b]$. Note that an elementary argument in distribution theory shows that a primitive U of a distribution on an interval (a, b) is uniquely determined up to an additive constant.

We shall now characterize all continuous functions u which are primitives of L_2 -functionals F on (a, b) . Consider a sequence $\{f_k\}$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. The preceding analysis and our last remark show that every primitive u of F has the form

$$u(x) = - \lim_{k \rightarrow \infty} \int_x^b f_k(t) dt + c \quad (3.11)$$

where c is a suitable constant. This implies by Schwarz's inequality that, for every subdivision $a \leq x_0 < x_1 < \dots < x_m \leq b$ of the interval $[a, b]$,

$$\begin{aligned} \sum_{j=1}^m \frac{|u(x_j) - u(x_{j-1})|^2}{x_j - x_{j-1}} &= \lim_{k \rightarrow \infty} \sum_{j=1}^m \frac{1}{x_j - x_{j-1}} \left| \int_{x_{j-1}}^{x_j} f_k(t) dt \right|^2 \\ &\leq \lim_{k \rightarrow \infty} \sum_{j=1}^m \int_{x_{j-1}}^{x_j} |f_k(t)|^2 dt \leq \lim_{k \rightarrow \infty} \|f_k\|^2 = \|F\|^2. \end{aligned} \quad (3.12)$$

On the other hand, assume that u is a continuous function satisfying the inequality

$$\sum_{j=1}^m \frac{|u(x_j) - u(x_{j-1})|^2}{x_j - x_{j-1}} \leq A \quad (3.13)$$

with a fixed constant A for all subdivisions $a \leq x_0 < x_1 < \cdots < x_m \leq b$ of the interval $[a, b]$. Let F be the functional derivative of the functional generated by u :

$$F\varphi = - \int u(x) \varphi'(x) dx. \quad (3.14)$$

By writing the integral in (3.14) as a Riemann sum, we obtain

$$F\varphi = - \lim_{\epsilon \rightarrow 0} \epsilon \sum_j u(j\epsilon) \varphi'(j\epsilon). \quad (3.15)$$

Note that the last sum consists of at most $[(b-a)/\epsilon] + 1$ nonzero terms where $[x]$ denotes the largest integer not exceeding x . By the mean value theorem,

$$\left| \frac{\varphi(j\epsilon) - \varphi((j-1)\epsilon)}{\epsilon} - \varphi'(j\epsilon) \right| \leq \epsilon \max |\varphi''(x)|. \quad (3.16)$$

Hence,

$$\begin{aligned} F\varphi &= - \lim_{\epsilon \rightarrow 0} \sum_j u(j\epsilon) [\varphi(j\epsilon) - \varphi((j-1)\epsilon)] \\ &= \lim_{\epsilon \rightarrow 0} \sum_j [u((j+1)\epsilon) - u(j\epsilon)] \varphi(j\epsilon) \\ &= \lim_{\epsilon \rightarrow 0} \sum_j \frac{u((j+1)\epsilon) - u(j\epsilon)}{\sqrt{\epsilon}} \cdot \sqrt{\epsilon} \varphi(j\epsilon). \end{aligned}$$

By applying Schwarz's inequality to the last sum and using (3.13), we obtain

$$\begin{aligned} |F\varphi| &\leq \overline{\lim}_{\epsilon \rightarrow 0} \left\{ \left[\sum_j \frac{|u((j+1)\epsilon) - u(j\epsilon)|^2}{\epsilon} \right] \cdot \epsilon \sum_j |\varphi(j\epsilon)|^2 \right\}^{1/2} \\ &\leq A^{1/2} \left[\int |\varphi(x)|^2 dx \right]^{1/2} = A^{1/2} \|\varphi\|. \end{aligned}$$

This inequality shows that F is a L_2 -functional satisfying the estimate

$$\|F\| \leq A^{1/2}. \quad (3.17)$$

By (3.12), the estimate (3.13) is satisfied with $A = \|F\|^2$. Thus we have proved:

THEOREM 3.1. *Let (a, b) be a bounded interval of the real axis. Every functional $F \in L_2(a, b)$ is a functional derivative of a linear functional F_u generated by a function $u \in C[a, b]$. Call a function u with this property a primitive of F . A function $u \in C[a, b]$ is a primitive of a functional $F \in L_2(a, b)$ if and only if a constant $A \geq 0$ exists such that the inequality (3.13) holds for every subdivision $a \leq x_0 < x_1 < \dots < x_m \leq b$ of $[a, b]$. For $F \in L_2(a, b)$, the smallest number A with this property is $A = \|F\|^2$. Hence the norm $\|F\|$ of every functional $F \in L_2(a, b)$ can be obtained from the primitive u of F by the formula*

$$\|F\| = \sup_{a \leq x_0 < x_1 < \dots < x_m \leq b} \left\{ \sum_{j=1}^m \frac{|u(x_j) - u(x_{j-1})|^2}{x_j - x_{j-1}} \right\}^{1/2}. \quad (3.18)$$

Theorem 3.1 is in complete agreement with the classical characterization of primitives of square-integrable functions which is based on the derivative concept of Lebesgue's theory and the theory of absolutely continuous functions (compare [5], p. 75).

Now we extend the first part of Theorem 3.1 to several variables and show:

THEOREM 3.2. *Assume that Ω is an open bounded subset of R_n and that $F \in L_2(\Omega)$. Then a function $u \in C(\bar{\Omega})$ exists such that (in the functional sense)*

$$F = \frac{\partial}{\partial x_1} \dots \frac{\partial}{\partial x_n} u = \frac{\partial^n}{\partial x_1 \dots \partial x_n} u. \quad (3.19)$$

Furthermore, u can be chosen in such a way that

$$|u(x)| \leq d(\Omega)^{n/2} \|F\| \quad \text{for} \quad x \in \Omega \quad (3.20)$$

where $d(\Omega)$ denotes the diameter of Ω .

PROOF. Choose points $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n)$ such that $b_i - a_i \leq d(\Omega)$ and $a_i < x_i < b_i$ ($i = 1, \dots, n$) for all points $x = (x_1, \dots, x_n)$ belonging to Ω and set

$$U\varphi = (-1)^n \lim_{k \rightarrow \infty} \int_{R_n} f_k(x) \left[\int_{a_1}^{x_1} \dots \int_{a_n}^{x_n} \varphi(\xi_1, \dots, \xi_n) d\xi_1 \dots d\xi_n \right] dx \quad (3.21)$$

where $f_k \in C_0^\infty(\Omega)$ and $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. Since

$$\left(\frac{\partial}{\partial x_1} \dots \frac{\partial}{\partial x_n} U \right) \varphi = (-1)^n U \left(\frac{\partial}{\partial x_1} \dots \frac{\partial}{\partial x_n} \varphi \right) \quad (3.22)$$

and since φ vanishes on every hyperplane $\xi_i = a_i$, we obtain in analogy to (3.6)

$$\frac{\partial}{\partial x_1} \dots \frac{\partial}{\partial x_n} U = F. \quad (3.23)$$

By interchanging the order of the integrations with respect to ξ_i and x_i in (3.21) successively for $i = 1, \dots, n$, we conclude that

$$U\varphi = \lim_{k \rightarrow \infty} \int u_k \varphi \, dx \quad (3.24)$$

where

$$u_k(x) = (-1)^n \int_{x_1}^{b_1} \cdots \int_{x_n}^{b_n} f_k(\xi_1, \dots, \xi_n) \, d\xi_1 \cdots d\xi_n. \quad (3.25)$$

By applying Schwarz's inequality n -times, we obtain for $x \in \bar{\Omega}$

$$\begin{aligned} |u_k(x)|^2 &= \left| \int_{x_1}^{b_1} \left[\int_{x_2}^{b_2} \cdots \int_{x_n}^{b_n} f_k(\xi_1, \dots, \xi_n) \, d\xi_2 \cdots d\xi_n \right] d\xi_1 \right|^2 \\ &\leq (b_1 - a_1) \int_{x_1}^{b_1} \left| \int_{x_2}^{b_2} \cdots \int_{x_n}^{b_n} f_k(\xi_1, \dots, \xi_n) \, d\xi_2 \cdots d\xi_n \right|^2 d\xi_1 \\ &\leq (b_1 - a_1) \cdots (b_n - a_n) \int_{x_1}^{b_1} \cdots \int_{x_n}^{b_n} |f_k(\xi)|^2 \, d\xi \leq d(\Omega)^n \|f_k\|^2 \end{aligned} \quad (3.26)$$

and in the same way

$$|u_k(x) - u_j(x)|^2 \leq d(\Omega)^n \|f_k - f_j\|^2 \quad \text{for } x \in \bar{\Omega}. \quad (3.27)$$

By (3.27), the sequence $\{u_k\}$ converges uniformly in $\bar{\Omega}$. Hence the limit function u is continuous, and (3.24) implies that

$$U\varphi = \int u \varphi \, dx. \quad (3.28)$$

Formula (3.19) is an immediate consequence of (3.23) and (3.28) while (3.20) follows from (3.26) and the limit relations $u_k(x) \rightarrow u(x)$ and $\|f_k\| \rightarrow \|F\|$ as $k \rightarrow \infty$. This concludes the proof of Theorem 3.2.

Theorem 3.2 has an interesting application to distribution theory. One of the basic theorems of distribution theory is L. Schwartz's result on the local behavior of distributions:

THEOREM 3.3. *Let F be a distribution in Ω . Then, for any compact subset K of Ω , there exist a positive integer m and a continuous function u defined on K such that*

$$F\varphi = (-1)^{mn} \int u(x) \left(\frac{\partial}{\partial x_1} \right)^m \cdots \left(\frac{\partial}{\partial x_n} \right)^m \varphi(x) \, dx \quad (3.29)$$

for every $\varphi \in C_0^\infty(\Omega)$ with $\text{supp } \varphi \subseteq K$.

Theorem 3.3 says that $F = (\partial/\partial x_1)^m \cdots (\partial/\partial x_n)^m u$ in every open subset Ω' of K , i.e., that F is locally of finite order. All known proofs of Theorem 3.3

are based on Lebesgue integration and employ the classical analogue of Theorem 3.2. By using Theorem 3.2, it is possible to replace the use of Lebesgue theory by the functional-analytic argument discussed here. We give a short outline of the proof. By a familiar argument of distribution theory, a real number $C > 0$ and a positive integer k can be chosen such that

$$|F\varphi| \leq C \text{Max} \left\{ \left| \left(\frac{\partial}{\partial x_1} \right)^{p_1} \cdots \left(\frac{\partial}{\partial x_n} \right)^{p_n} \varphi(x) \right| : x \in K, p_1 + \cdots + p_n \leq k \right\}$$

for every $\varphi \in C_0^\infty(\Omega)$ with $\text{supp } \varphi \subseteq K$. Set

$$D(K) = \{\varphi \in C_0^\infty(R_n) : \text{supp } \varphi \subseteq K\}, \quad (3.30)$$

$$A\varphi = \left(\frac{\partial}{\partial x_1} \cdots \frac{\partial}{\partial x_n} \right)^{k+1} \varphi, \quad (3.31)$$

$$\Psi = \{\psi = A\varphi : \varphi \in D(K)\}. \quad (3.32)$$

It is easy to show (compare [10], p. 101) that the mapping A from $D(K)$ onto Ψ is one-to-one and that the linear functional L on Ψ defined by

$$L\psi = F(A^{-1}\psi) \quad \text{for} \quad \psi \in \Psi \quad (3.33)$$

is bounded with respect to the norm (1.2). Let $\overline{\Psi}$ be the set of all functionals $F \in L_2(\Omega)$ with the property that a sequence $\{f_j\}$ in Ψ exists such that $\|F - f_j\| \rightarrow 0$ as $j \rightarrow \infty$. It follows as in the proof of Lemma 2.1 that $\overline{\Psi}$ is a Hilbert space. Since Ψ is dense in $\overline{\Psi}$, L can be uniquely extended to a bounded linear functional on $\overline{\Psi}$. By the representation theorem of F. Riesz and by Lemma 2.3, there exists a functional $H \in \overline{\Psi}$ such that

$$F(A^{-1}\psi) = L\psi = (\psi, H) \quad \text{for every} \quad \psi \in \Psi. \quad (3.34)$$

Since $H \in \overline{\Psi}$ we can choose a sequence $\{h_k\}$ in Ψ such that $\|H - h_k\|_1 \rightarrow 0$ as $k \rightarrow \infty$. Note that h_k belongs to Ψ . Hence there exists a $G \in \overline{\Psi}$ such that $\|G - h_k\|_1 \rightarrow 0$ as $k \rightarrow \infty$ (3.34) and Lemma 2.3 yield

$$F(A^{-1}\psi) = \lim_{k \rightarrow \infty} (\psi, h_k) = \lim_{k \rightarrow \infty} (h_k, \bar{\psi}) = (G, \bar{\psi}) = G\psi.$$

By applying this formula to $\psi = A\varphi$, we obtain

$$F\varphi = G(A\varphi) = G \left(\left(\frac{\partial}{\partial x_1} \cdots \frac{\partial}{\partial x_n} \right)^{k+1} \varphi \right) \quad \text{for every} \quad \varphi \in D(K). \quad (3.35)$$

Choose a bounded open set Ω' with $K \subset \Omega' \subseteq \Omega$. The restriction G^r of G to $C_0^\infty(\Omega')$ belongs to $L_2(\Omega')$. In fact, since $G \in \overline{\Psi}$ and $K \subset \Omega'$, there exists a sequence $\{g_j\}$ in $\Psi \subset C_0^\infty(\Omega')$ such that

$$\|G^r - g_j\|_{L_2(\Omega')} \leq \|G - g_j\|_{L_2(\Omega)} \rightarrow 0 \quad \text{as} \quad j \rightarrow \infty. \quad (3.36)$$

By applying Theorem 3.2 to the bounded domain Ω' and the functional $(-1)^{n(k+1)} G^r$, we conclude that a function $u \in C(\bar{\Omega}')$ exists such that

$$G\chi = G^r\chi = (-1)^{n(k+2)} \int u \frac{\partial}{\partial x_1} \cdots \frac{\partial}{\partial x_n} \chi \, dx \quad \text{for every } \chi \in C_0^\infty(\Omega'). \quad (3.37)$$

Set $m = k + 2$ and $\chi = A\varphi = (\partial/\partial x_1 \cdots \partial/\partial x_n)^{k+1} \varphi$ where $\varphi \in D(K)$. Then (3.29) follows immediately from (3.35) and (3.37). This concludes the proof of Theorem 3.3. In particular, our considerations show that the main result on the local behavior of distributions can be obtained completely within the framework of distribution theory, without applying any results of Lebesgue's theory.

4. SOBOLEV SPACES

In this section we consider linear spaces consisting of L_2 -functionals with the property that their derivatives up to a certain order are also L_2 -functionals. Spaces of this type were first considered by S. L. Sobolev in the framework of classical Lebesgue theory ([7], [8]) and found important applications in the Hilbert space theory of differential operators (see, e.g. [1] and [8]).

We employ the usual notation for higher order derivatives. Let $x = (x_1, \dots, x_n)$ be a point of the n -dimensional space R_n and $p = (p_1, \dots, p_n)$ a multi-index, with $p_1, \dots, p_n$ denoting non-negative integers. We set

$$x^p = x_1^{p_1} \cdots x_n^{p_n}, \quad D^p = \left(\frac{\partial}{\partial x_1} \right)^{p_1} \cdots \left(\frac{\partial}{\partial x_n} \right)^{p_n}, \\ |p| = p_1 + \cdots + p_n, \quad x^{(0, \dots, 0)} = 1, \quad D^{(0, \dots, 0)} = 1. \quad (4.1)$$

DEFINITION 4.1. $H_m(\Omega)$ is the linear space of all linear functionals F on $C_0^\infty(\Omega)$ with the property that F and all derivatives $D^p F$ of order $|p| \leq m$ belong to $L_2(\Omega)$.

We introduce in $H_m(\Omega)$ an inner product by

$$(F, G)_m = \sum_{0 \leq |p| \leq m} (D^p F, D^p G) \quad (4.2)$$

where the inner products on the right-hand side are defined as in Section 2. The related norm is given by

$$\|F\|_m = (F, F)_m^{1/2} = \left[\sum_{|p| \leq m} \|D^p F\|^2 \right]^{1/2}. \quad (4.3)$$

THEOREM 4.1. $H_m(\Omega)$ forms a Hilbert space with respect to the inner product (4.2).

PROOF. Consider a sequence $\{F_k\}$ in $H_m(\Omega)$ such that $\|F_k - F_j\|_m \rightarrow 0$ as $k, j \rightarrow \infty$. By (4.3) the sequences $\{D^p F_k\}$, $0 \leq |p| \leq m$, converge in $L_2(\Omega)$. Hence, by the completeness of $L_2(\Omega)$, there exist functionals $F^p \in L_2(\Omega)$ such that $\|D^p F_k - F^p\| \rightarrow 0$ as $k \rightarrow \infty$. Set $F = F^{(0, \dots, 0)}$. We show that $F^p = D^p F$. By (3.1) we have $(D^p F)\varphi = (-1)^{|p|} F(D^p \varphi)$. Thus we obtain for $\varphi \in C_0^\infty(\Omega)$ and $|p| \leq m$

$$\begin{aligned} |F^p \varphi - (D^p F)\varphi| &= |F^p \varphi - (-1)^{|p|} F(D^p \varphi)| \\ &\leq |F^p \varphi - (D^p F_k)\varphi| + |(D^p F_k)\varphi - (-1)^{|p|} F(D^p \varphi)| \\ &= |(F^p - D^p F_k)\varphi| + |(F_k - F)D^p \varphi| \\ &\leq \|F^p - D^p F_k\| \cdot \|\varphi\| + \|F_k - F\| \cdot \|D^p \varphi\| \rightarrow 0 \\ &\quad \text{as } k \rightarrow \infty \end{aligned}$$

and hence $F^p \varphi = (D^p F)\varphi$ for every $\varphi \in C_0^\infty(\Omega)$. This implies that $D^p F = F^p \in L_2(\Omega)$ if $|p| \leq m$. Thus we have shown that $F \in H_m(\Omega)$. Furthermore, we have $\|D^p F_k - D^p F\| = \|D^p F_k - F^p\| \rightarrow 0$ and hence, by (4.3), $\|F_k - F\|_m \rightarrow 0$ as $k \rightarrow \infty$. This shows that $H_m(\Omega)$ is complete with respect to the norm (4.3) which concludes the proof of Theorem 4.1.

Note that $C_0^\infty(\Omega)$ is a subspace of $H_m(\Omega)$. The following linear spaces $\dot{H}_m(\Omega)$ are useful for the discussion of boundary value problems with Dirichlet boundary data:

DEFINITION 4.2. $\dot{H}_m(\Omega)$ is the linear subspace of $H_m(\Omega)$ consisting of all functionals $F \in H_m(\Omega)$ with the property that a sequence $\{f_k\}$ exists in $C_0^\infty(\Omega)$ such that $\|F - f_k\|_m \rightarrow 0$ as $k \rightarrow \infty$.

The same argument as in the proof of Lemma 2.1 shows that $\dot{H}_m(\Omega)$ is complete with respect to the norm (4.3). Hence $\dot{H}_m(\Omega)$ is also a Hilbert space with the inner product (4.2). We show:

LEMMA 4.1.

$$C_0^m(\Omega) \subset \dot{H}_m(\Omega).$$

PROOF. Assume that $u \in C_0^m(\Omega)$ and define u_k by (2.17). Note that $\delta = \text{dist}(\text{supp } u, \partial\Omega) > 0$ and that $u_k \in C_0^\infty(\Omega)$ if $k > 1/\delta$. (2.17) implies that for $|p| \leq m$

$$D^p u_k(x) = \int D_x^p u \left(x + \frac{y}{k} \right) h(y) dy. \quad (4.4)$$

This shows, by the argument following (2.18) and by (2.20), that

$$\|D^p u - D^p u_k\|^2 = \int |D^p u - D^p u_k|^2 dx \rightarrow 0 \quad \text{for} \quad |p| \leq m. \quad (4.5)$$

Hence, u can be approximated in $H_m(\Omega)$ by the functions $u_k \in C_0^\infty(\Omega)$. This completes the proof of Lemma 4.1.

Our next theorem is related to the classical integration by parts formula:

THEOREM 4.2. *Assume that $F \in L_2(\Omega)$, $D^p F \in L_2(\Omega)$, and $G \in \dot{H}_{|p|}(\Omega)$. Then*

$$(D^p F, G) = (-1)^{|p|} (F, D^p G). \quad (4.6)$$

In particular, (4.6) holds if $F \in H_m(\Omega)$, $G \in \dot{H}_m(\Omega)$ and $|p| \leq m$.

PROOF. Since $G \in \dot{H}_{|p|}(\Omega)$, there exists a sequence $\{g_k\}$ in $C_0^\infty(\Omega)$ such that $\|G - g_k\| \rightarrow 0$ and $\|D^p G - D^p g_k\| \rightarrow 0$ as $k \rightarrow \infty$. Hence, Lemma 2.3 implies that

$$\begin{aligned} (F, D^p G) &= \lim_{k \rightarrow \infty} (F, D^p g_k) = \lim_{k \rightarrow \infty} F(D^p \bar{g}_k) \\ &= (-1)^{|p|} \lim_{k \rightarrow \infty} (D^p F) \bar{g}_k = (-1)^{|p|} \lim_{k \rightarrow \infty} (D^p F, g_k) \\ &= (-1)^{|p|} (D^p F, G). \end{aligned}$$

We conclude this section with some remarks on the functionals F^* and F^τ introduced at the end of Section 2.

LEMMA 4.2. *Consider two open subsets Ω, Ω' of R_n with $\Omega \subset \Omega'$ and assume that $F \in \dot{H}_m(\Omega)$. Let F^* and $(D^p F)^*$, $|p| \leq m$, be the norm-invariant extensions of F and $D^p F$ to $L_2(\Omega')$ defined by (2.24). Then*

$$D^p F^* = (D^p F)^*. \quad (4.7)$$

Furthermore, $F^ \in \dot{H}_m(\Omega')$ and*

$$\|F^*\|_{m, \Omega'} = \|F\|_{m, \Omega}. \quad (4.8)$$

PROOF. Choose a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ and that $\|F - f_k\|_{m, \Omega} \rightarrow 0$ as $k \rightarrow \infty$. (3.1) and (2.24) imply that for $|p| \leq m$ and $\varphi \in C_0^\infty(\Omega')$

$$\begin{aligned} (D^p F^*) \varphi &= (-1)^{|p|} F^*(D^p \varphi) = (-1)^{|p|} \lim_{k \rightarrow \infty} \int f_k D^p \varphi \, dx \\ &= \lim_{k \rightarrow \infty} \int \varphi D^p f_k \, dx = (D^p F)^* \varphi. \end{aligned}$$

This proves (4.7). Hence we have, by (2.26),

$$D^p F^* \in L_2(\Omega') \quad \text{and} \quad \|D^p F^*\|_{L_2(\Omega')} = \|D^p F\|_{L_2(\Omega)}$$

which yields (4.8). By applying (4.8) to $F - f_k$, we obtain,

$$\|F^* - f_k\|_{m, \Omega'} = \|F - f_k\|_{m, \Omega} \rightarrow 0$$

as $k \rightarrow \infty$. Hence $F^* \in \dot{H}_m(\Omega')$. This concludes the proof of Lemma 4.2.

LEMMA 4.3. *Assume that $\Omega \subset \Omega'$ and that $F \in H_m(\Omega')$. Let F^r and $(D^p F)^r$ be the restrictions of F and $D^p F$ to $C_0^\infty(\Omega)$ defined by (2.22). Then*

$$D^p F^r = (D^p F)^r. \quad (4.9)$$

In particular, $F^r \in H_m(\Omega)$ and

$$\|F^r\|_{m, \Omega} \leq \|F\|_{m, \Omega'}. \quad (4.10)$$

PROOF. Assume that $\varphi \in C_0^\infty(\Omega)$. Then (4.9) follows from

$$(D^p F^r) \varphi = (-1)^{|p|} F^r(D^p \varphi) = (-1)^{|p|} F(D^p \varphi) = (D^p F) \varphi$$

while (4.10) is a consequence of (2.23) and (4.9).

It is convenient to use the notation $F \in H_m(\Omega)$ if F is a linear functional on $C_0^\infty(\Omega')$, with $\Omega \subset \Omega'$, and if the restriction of F to $C_0^\infty(\Omega)$ belongs to $H_m(\Omega)$. Furthermore, we say that $F_1 = F_2$ in Ω if the restrictions of F_1 and F_2 to $C_0^\infty(\Omega)$ coincide.

5. PRODUCTS WITH CONTINUOUS FUNCTIONS

In this section we define the product of a L_2 -functional with a continuous function. Let us recall that the product gF of a distribution F in Ω with a function $g \in C_0^\infty(\Omega)$ is defined as the linear functional on $C_0^\infty(\Omega)$ given by

$$(gF) \varphi = F(g\varphi) \quad \text{for} \quad \varphi \in C_0^\infty(\Omega). \quad (5.1)$$

Assume that $F \in L_2(\Omega)$. Then Lemma 2.3 implies that (5.1) can be written in the form

$$(gF) \varphi = (F, \bar{g} \cdot \bar{\varphi}) \quad \text{for} \quad \varphi \in C_0^\infty(\Omega). \quad (5.2)$$

Note that the right-hand side is also defined if $g \in C(\Omega)$. In fact, if g is continuous in Ω , then $g \cdot \varphi$ belongs to $C_0(\Omega)$ for every $\varphi \in C_0^\infty(\Omega)$. Hence

Lemma 2.5 shows that $g \cdot \varphi \in C(\Omega) \cap L_2(\Omega)$ so that the inner product in (5.2) is well defined. This remark motivates the following definition:

DEFINITION 5.1. Assume that $g \in C(\Omega)$ and $F \in L_2(\Omega)$. Then the product functional gF is defined by (5.2).

We show:

LEMMA 5.1. *If $g \in C_b(\Omega)^1$ and $F \in L_2(\Omega)$, then also $gF \in L_2(\Omega)$. Furthermore, we have*

$$\|gF\| \leq \|g\|_\infty \cdot \|F\| \quad (5.3)$$

where

$$\|g\|_\infty = \sup_{x \in \Omega} |g(x)|. \quad (5.4)$$

Note that $C_b(\Omega)$ is a Banach space with respect to the norm (5.4)

PROOF OF LEMMA 5.1. By Lemma 2.5 and (5.4), we have

$$\|g \cdot \varphi\| = \left[\int |g \cdot \varphi|^2 dx \right]^{1/2} \leq \|g\|_\infty \cdot \|\varphi\| \quad (5.5)$$

and hence, by (5.2) and Schwarz's inequality,

$$|(gF) \varphi| \leq \|F\| \cdot \|g\varphi\| \leq \|g\|_\infty \cdot \|F\| \cdot \|\varphi\| \quad (5.6)$$

for every $\varphi \in C_0^\infty(\Omega)$. This implies 5.3.

Our next Lemma shows that the product gF reduces to the usual product $g \cdot u$ of two functions if $F = F_u$ is generated by a continuous function u .

LEMMA 5.2. *Assume that $g \in C(\Omega)$ and $u \in C(\Omega) \cap L_2(\Omega)$. Then the functionals F_u and $F_{g \cdot u}$ generated by u and $g \cdot u$ are related by*

$$F_{g \cdot u} = gF_u. \quad (5.7)$$

If, in addition, $g \in C_b(\Omega)$ then $g \cdot u \in C(\Omega) \cap L_2(\Omega)$.

PROOF. Consider a function $\varphi \in C_0^\infty(\Omega)$. Since $\text{supp } \varphi$ is a compact subset of Ω , it is possible to choose a function $\zeta \in C_0^\infty(\Omega)$ such that

$$\zeta(x) = 1 \quad \text{if} \quad x \in \text{supp } \varphi \quad (5.8)$$

¹ See Section 0 for the definition of $C_b(\Omega)$ and $C_b^m(\Omega)$.

(see, e.g., [2], Theorem 1.2.2). Since $g \cdot \zeta \in C_0(\Omega)$, we conclude from Lemma 2.5 and Theorem 2.1 that $g \cdot \zeta \in L_2(\Omega)$ and that a sequence $\{\psi_k\}$ exists in $C_0^\infty(\Omega)$ such that

$$\|g\zeta - \psi_k\| = \left[\int |g\zeta - \psi_k|^2 dx \right]^{1/2} \rightarrow 0 \quad \text{as} \quad k \rightarrow \infty. \quad (5.9)$$

Since $\|\varphi(g\zeta - \psi_k)\| \leq \|\varphi\|_\infty \cdot \|g\zeta - \psi_k\|$, we have also

$$\|g\zeta\varphi - \psi_k\varphi\| \rightarrow 0 \quad \text{as} \quad k \rightarrow \infty. \quad (5.10)$$

Note that $\zeta \cdot \varphi = \varphi$ because of (5.8). Hence (5.2), (5.10) and Lemma 2.3 imply that

$$\begin{aligned} (gF_u)\varphi &= (gF_u)(\zeta\varphi) = (F_u, \overline{g \cdot \zeta \cdot \varphi}) = \lim_{k \rightarrow \infty} (F_u, \overline{\psi_k\varphi}) \\ &= \lim_{k \rightarrow \infty} F_u(\psi_k\varphi) = \lim_{k \rightarrow \infty} \int u\psi_k\varphi dx = \int ug\zeta\varphi dx. \end{aligned} \quad (5.11)$$

The last equality follows from (5.10) and the estimate

$$\left| \int u(\psi_k\varphi - g\zeta\varphi) dx \right| \leq \|g\zeta\varphi - \psi_k\varphi\| \cdot \sup_{x \in \text{supp } \varphi} |u(x)|.$$

Since $\zeta \cdot \varphi = \varphi$, (5.11) yields

$$(gF_u)\varphi = \int ug\varphi dx = F_{g \cdot u}\varphi. \quad (5.12)$$

This relation holds for every $\varphi \in C_0^\infty(\Omega)$. Thus we have proved (5.7). If $g \in C_b(\Omega)$ then Lemma 5.1 implies that gF_u and hence, by (5.7), $F_{g \cdot u}$ belongs to $L_2(\Omega)$. But this means that $g \cdot u \in C(\Omega) \cap L_2(\Omega)$. This completes the proof of Lemma 5.2.

The following rule is often useful:

LEMMA 5.3. Assume that $F, G \in L_2(\Omega)$ and $u \in C_b(\Omega)$. Then

$$(uF, G) = (F, \bar{u}G). \quad (5.13)$$

PROOF. Choose sequences $\{f_k\}, \{g_k\}$ in $C_0^\infty(\Omega)$ such that $\|F - f_k\| \rightarrow 0$ and $\|G - g_k\| \rightarrow 0$ as $k \rightarrow \infty$. By (5.3) we have also $\|uF - uf_k\| \rightarrow 0$ and $\|\bar{u}G - \bar{u}g_k\| \rightarrow 0$ as $k \rightarrow \infty$. Hence Lemma 2.5 implies that

$$\begin{aligned} (uF, G) &= \lim_{k \rightarrow \infty} (uf_k, g_k) = \lim_{k \rightarrow \infty} \int uf_k \bar{g}_k dx \\ &= \lim_{k \rightarrow \infty} \int f_k \overline{ug_k} dx = \lim_{k \rightarrow \infty} (f_k, \bar{u}g_k) = (F, \bar{u}G). \end{aligned}$$

This concludes the proof of Lemma 5.3.

Now we study the differentiation of product functionals. We use the following notation for multi-indices $p = (p_1, \dots, p_n)$ and $r = (r_1, \dots, r_n)$:

$$\begin{aligned} p! &= p_1! \cdot \dots \cdot p_n!, \\ p - r &= (p_1 - r_1, \dots, p_n - r_n), \\ r \leq p &\quad \text{if} \quad r_1 \leq p_1, \dots, r_n \leq p_n, \\ \binom{p}{r} &= \frac{p!}{r!(p-r)!} \quad \text{for} \quad r \leq p. \end{aligned} \quad (5.14)$$

We show:

THEOREM 5.1. *Assume that $D^r F \in L_2(\Omega)$ if $r \leq p$ and that $g \in C^{|\mathbf{p}|}(\Omega)$. Then the derivative $D^p(gF)$ of the product functional gF is given by*

$$D^p(gF) = \sum_{0 \leq r \leq p} \binom{p}{r} (D^r g) \cdot (D^{p-r} F). \quad (5.15)$$

In particular, if $F \in H_m(\Omega)$ and $g \in C_b^m(\Omega)$ then $gF \in H_m(\Omega)$.

PROOF. First we consider the case $m = 1$. Assume that $F \in L_2(\Omega)$, $(\partial/\partial x_i)F \in L_2(\Omega)$, $g \in C^1(\Omega)$, and $\varphi \in C_0^\infty(\Omega)$. By (3.1) and (5.2), we obtain

$$\begin{aligned} \left[\frac{\partial}{\partial x_i} (gF) \right] \varphi &= - (gF) \left(\frac{\partial}{\partial x_i} \varphi \right) = - \left(F, g \overline{\frac{\partial}{\partial x_i} \varphi} \right) \\ &= - \left(F, \frac{\partial}{\partial x_i} \overline{(g \cdot \varphi)} \right) + \left(F, \varphi \overline{\frac{\partial}{\partial x_i} g} \right). \end{aligned} \quad (5.16)$$

Since $g \cdot \varphi \in C_0^1(\varphi)$, Lemma 4.1 implies that $g \cdot \varphi \in \dot{H}_1(\Omega)$. Hence, by Theorem 4.2, we have

$$- \left(F, \frac{\partial}{\partial x_i} \overline{(g \cdot \varphi)} \right) = \left(\frac{\partial}{\partial x_i} F, \overline{g \cdot \varphi} \right). \quad (5.17)$$

By inserting (5.17) into (5.16) and applying (5.2), we obtain

$$\left[\frac{\partial}{\partial x_i} (gF) \right] \varphi = \left(g \frac{\partial}{\partial x_i} F \right) \varphi + \left(\frac{\partial g}{\partial x_i} F \right) \varphi \quad (5.18)$$

for every $\varphi \in C_0^\infty(\Omega)$ and hence

$$\frac{\partial}{\partial x_i} (gF) = g \frac{\partial}{\partial x_i} F + \frac{\partial g}{\partial x_i} F. \quad (5.19)$$

Thus (5.15) is valid if $m = 1$. The general case follows easily by induction. If $F \in H_m(\Omega)$ and $g \in C_0^m(\Omega)$ then (5.15) and Lemma 5.1 imply that $D^p(gF) \in L_2(\Omega)$ for $|p| \leq m$. Hence gF belongs to $H_m(\Omega)$. This concludes the proof of Theorem 5.1.

COROLLARY TO THEOREM 5.1. *Assume that $F \in \dot{H}_m(\Omega)$ and $g \in C_0^m(\Omega)$. Then $gF \in \dot{H}_m(\Omega)$.*

PROOF. By Definition 4.2, there exists a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ such that $\|F - f_k\|_m \rightarrow 0$ as $k \rightarrow \infty$. By Theorem 5.1 we have for $|p| \leq m$

$$D^p(gF) - D^p(gf_k) = \sum_{0 \leq r \leq p} \binom{p}{r} D^r g \cdot D^{p-r}(F - f_k) \quad (5.20)$$

and hence, by (5.3), $\|D^p(gF) - D^p(gf_k)\| \rightarrow 0$ as $k \rightarrow \infty$. This shows that $\|gF - gf_k\|_m \rightarrow 0$ as $k \rightarrow \infty$. Since $gf_k \in C_0^m(\Omega)$ we can choose, by Lemma 4.1, a sequence $\{\psi_k\}$ in $C_0^\infty(\Omega)$ such that $\|gf_k - \psi_k\|_m < 1/k$ for every k . By the triangle inequality we have $\|gF - \psi_k\|_m \rightarrow 0$ as $k \rightarrow \infty$. Hence $gF \in \dot{H}_m(\Omega)$ which proves the Corollary.

6. FOURIER TRANSFORMS

In this section we define the Fourier transform $\hat{F}$ of a functional $F \in L_2(R_n)$. Consider the linear space S of all functions $\varphi \in C^\infty(R_n)$ with the property that

$$\sup_{x \in R_n} \{ |x^p| \cdot |D^q \varphi(x)| \} < \infty \quad (6.1)$$

for every pair of multi-indices $p = (p_1, \dots, p_n)$, $q = (q_1, \dots, q_n)$. The Fourier transform $\hat{\varphi}$ of a function $\varphi \in S$ is defined by

$$\hat{\varphi}(\xi) = (2\pi)^{-n/2} \int e^{ix \cdot \xi} \varphi(x) dx \quad (6.2)$$

where $x \cdot \xi = x_1 \xi_1 + \dots + x_n \xi_n$. It is easy to show that $\varphi \rightarrow \hat{\varphi}$ is a one-to-one linear mapping of S onto S and that Parseval's relation

$$\int |\hat{\varphi}|^2 d\xi = \int |\varphi|^2 dx \quad (6.3)$$

and Fourier's inversion formula

$$\varphi(x) = \hat{\hat{\varphi}}(-x) = (2\pi)^{-n/2} \int e^{-ix \cdot \xi} \hat{\varphi}(\xi) d\xi \quad (6.4)$$

hold if $\varphi \in S$. A proof can be found, for example, in [2], Section 1.7. Note that all integrals in (6.2-4) exist because of (6.1).

Every function $\varphi \in S$ can be approximated by a sequence $\{\varphi_k\}$ in $C_0^\infty(R_n)$ such that

$$\int |\varphi - \varphi_k|^2 dx \rightarrow 0 \quad \text{as} \quad k \rightarrow \infty. \quad (6.5)$$

In fact, choose a sequence $\{\chi_k\}$ such that

$$\chi_k \in C_0^\infty(R_n), \quad 0 \leq \chi_k(x) \leq 1, \quad \chi_k(x) = 1 \quad \text{if} \quad |x| \leq k \quad (6.6)$$

(for example, by applying [2], Theorem 1.2.2) and set $\varphi_k = \varphi \chi_k$. Then (6.5) follows from

$$\int |\varphi - \varphi_k|^2 dx = \int_{|x| > k} |(1 - \chi_k) \varphi|^2 dx \leq \int_{|x| > k} |\varphi|^2 dx \rightarrow 0 \quad \text{as} \quad k \rightarrow \infty.$$

The second part of Lemma 2.5 implies that every function $u \in S$ generates a functional $F_u \in L_2(R_n)$ and that the relation (2.21) holds for $u, v \in S$. Hence S can be considered as a subspace of $L_2(R_n)$ by identifying u with the linear functional F_u . Furthermore, (2.21) and (6.3) show that the mapping $\varphi \rightarrow \hat{\varphi}$ is an isometry of S onto S with respect to the inner product in the Hilbert space $L_2(R_n)$. By Theorem 2.1, S is dense in $L_2(R_n)$ since $C_0^\infty(R_n) \subset S$.

Since S is dense in $L_2(R_n)$, the isometry $\varphi \rightarrow \hat{\varphi}$ of S onto S can be uniquely extended to an isometry $F \rightarrow \hat{F}$ of $L_2(R_n)$ onto $L_2(R_n)$. In fact, consider an element $F \in L_2(R_n)$ and a sequence $\{f_k\}$ in $C_0^\infty(R_n)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. Since $\|f_k - f_j\| = \|f_k - f_j\|$, the sequence $\{\hat{f}_k\}$ converges in $L_2(R_n)$. Since $L_2(R_n)$ is complete, there exists a $\hat{F} \in L_2(R_n)$ such that $\|\hat{F} - \hat{f}_k\| \rightarrow 0$ as $k \rightarrow \infty$. Let $\{f_k'\}$ be another sequence in $C_0^\infty(R_n)$ with $\|F - f_k'\| \rightarrow 0$ as $k \rightarrow \infty$. Since $\|f_k - f_k'\| = \|f_k - f_k'\| \rightarrow 0$ as $k \rightarrow \infty$, the sequences $\{\hat{f}_k\}$ and $\{\hat{f}_k'\}$ converge to the same limit in $L_2(R_n)$. This shows that $\hat{F}$ is independent of the choice of the sequence $\{f_k\}$. Since $\|F - f_k\| \rightarrow 0$ and $\|\hat{F} - \hat{f}_k\| \rightarrow 0$ as $k \rightarrow \infty$, we have

$$\|\hat{F}\| = \lim_{k \rightarrow \infty} \|\hat{f}_k\| = \lim_{k \rightarrow \infty} \|f_k\| = \|F\|. \quad (6.7)$$

Hence the mapping $F \rightarrow \hat{F}$ is an isometry. To show that $F \rightarrow \hat{F}$ maps $L_2(R_n)$ onto $L_2(R_n)$, we construct for any given $G \in L_2(R_n)$ an element $F \in L_2(R_n)$ such that $\hat{F} = G$. Choose a sequence $\{g_k\}$ in $C_0^\infty(R_n)$ such that $\|G - g_k\| \rightarrow 0$ as $k \rightarrow \infty$ and set $f_k(x) = \hat{g}_k(-x)$. Then f_k belongs to S , and (6.2) and (6.4) imply that $\hat{f}_k = g_k$. The sequence $\{f_k\}$ converges in $L_2(R_n)$ because $\|f_k - f_j\| = \|\hat{g}_k - \hat{g}_j\| = \|g_k - g_j\|$. Denote the limit of the sequence $\{f_k\}$ by F . (6.7) implies that $\|\hat{F} - \hat{f}_k\| \rightarrow 0$ as $k \rightarrow \infty$. On the other hand, we have

$\|G - f_k\| \rightarrow 0$ since $f_k = g_k$. Hence we obtain $\hat{F} = G$. Finally we show that $F \rightarrow \hat{F}$ is the only linear isometric mapping which extends the isometry $\varphi \rightarrow \hat{\varphi}$ ($\varphi \in S$) to $L_2(R_n)$. Assume that $\hat{F}$ is another such mapping. Then $\|F - f_k\| \rightarrow 0$ implies $\|\hat{F} - f_k\| \rightarrow 0$ and hence $\hat{F} = F$. Thus we have proved:

THEOREM 6.1. *The set S of all functions $\varphi \in C^\infty(R_n)$ with the property (6.1) is a linear subspace of $L_2(R_n)$. There exists one and only one linear isometric mapping $F \rightarrow \hat{F}$ from $L_2(R_n)$ onto $L_2(R_n)$ which extends the mapping $\varphi \rightarrow \hat{\varphi}$ defined on S by (6.2). For every $F \in L_2(R_n)$ and every sequence $\{f_k\}$ in $C_0^\infty(R_n)$ with $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$, the image $\hat{F}$ of F can be obtained by the limit relation $\|\hat{F} - f_k\| \rightarrow 0$ as $k \rightarrow \infty$.*

This leads to the following definition:

DEFINITION 6.1. Let $F \rightarrow \hat{F}$ be the isometry of $L_2(R_n)$ onto $L_2(R_n)$ described in Theorem 6.1. Then the image element $\hat{F}$ of an element $F \in L_2(R_n)$ is called the Fourier transform of F .

By formula (6.7), Parseval's relation

$$\|\hat{F}\| = \|F\| \quad (6.8)$$

holds for every $F \in L_2(R_n)$. Since

$$(F, G) = \frac{1}{4}(\|F + G\|^2 - \|F - G\|^2 + i\|F + iG\|^2 - i\|F - iG\|^2), \quad (6.9)$$

Parseval's relation (6.8) can also be written in the form

$$(\hat{F}, \hat{G}) = (F, G) \quad \text{for} \quad F, G \in L_2(R_n). \quad (6.10)$$

In order to extend the inversion formula (6.4) to $L_2(R_n)$, we introduce for every $F \in L_2(R_n)$ the linear functional $\tilde{F}$ defined by

$$\tilde{F}\varphi = F\tilde{\varphi} \quad \text{with} \quad \tilde{\varphi}(x) = \varphi(-x), \quad \varphi \in C_0^\infty(R_n). \quad (6.11)$$

Since $\tilde{\tilde{\varphi}} = \varphi$ and $\|\tilde{\varphi}\| = \|\varphi\|$, we have $\tilde{\tilde{F}} \in L_2(R_n)$, $\tilde{\tilde{F}} = F$ and $\|\tilde{F}\| = \|F\|$. Furthermore, if $F = F_u$ is generated by a continuous function u , then the relation

$$\tilde{F}_u\varphi = F_u\tilde{\varphi} = \int u(x)\varphi(-x)dx = \int u(-x)\varphi(x)dx = \int \tilde{u}\varphi dx = F_{\tilde{u}}\varphi \quad (6.12)$$

implies that $\tilde{F}_u$ is generated by $\tilde{u}(x) = u(-x)$. Hence the notations $\tilde{F}$ and $\tilde{u}$ are compatible.

Assume that $F \in L_2(R_n)$ and choose a sequence $\{f_k\}$ in $C_0^\infty(R_n)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. Since the mappings $F \rightarrow \hat{F}$ and $F \rightarrow \tilde{F}$ are linear

and isometric, we have $\|\hat{F} - \hat{f}_k\| \rightarrow 0$ and $\|\check{F} - \check{f}_k\| \rightarrow 0$ as $k \rightarrow \infty$. On the other hand, (6.4) implies that $\hat{f}_k = \check{f}_k$. Thus we obtain

$$\hat{\hat{F}} = \check{\check{F}} \quad (6.13)$$

for every $F \in L_2(R_n)$. (6.13) can be considered as the generalization of the inversion formula (6.4) to $L_2(R_n)$.

We conclude this section with a discussion of the relation between Definition 6.1 and the definition of the Fourier transform of a tempered distribution. Recall that a tempered distribution F is defined as a linear functional on S with the property that $F\varphi_k \rightarrow 0$ for all sequences $\{\varphi_k\}$ in $C_0^\infty(\Omega)$ such that (i) $\{D^p\varphi_k\}$ converges uniformly to zero in R_n and (ii)

$$\sup\{|x|^p \cdot |D^q\varphi_k(x)| : x \in R_n, k = 1, 2, \dots\} < \infty \quad (6.14)$$

for every pair of multi-indices p, q . The Fourier transform $\hat{F}$ of a tempered distribution F is defined as the linear functional on S given by

$$\hat{F}\varphi = F\hat{\varphi} \quad \text{for} \quad \varphi \in S. \quad (6.15)$$

Since $C_0^\infty(R_n)$ is dense in S with respect to the L_2 -norm, every functional $F \in L_2(R_n)$ can be uniquely extended to a bounded linear functional F^* on S . Properties (i) and (ii) imply that $\|\varphi_k\| \rightarrow 0$ and hence $F^*\varphi_k \rightarrow 0$. Thus F^* is a tempered distribution. Consider a sequence $\{f_k\}$ in $C_0^\infty(R_n)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. Then F^* is given by

$$F^*\varphi = \lim_{k \rightarrow \infty} \int f_k \varphi \, dx = \lim_{k \rightarrow \infty} (f_k, \bar{\varphi}) = (F, \bar{\varphi}) \quad \text{for every} \quad \varphi \in S. \quad (6.16)$$

Now we determine the extension $(\hat{F})^*$ of the Fourier transform $\hat{F}$ of F . Since $\|\hat{F} - \hat{f}_k\| \rightarrow 0$ as $k \rightarrow \infty$, we obtain by using the elementary identity

$$\int \varphi \hat{\psi} \, dx = \int \hat{\varphi} \psi \, dx \quad \text{for} \quad \varphi, \psi \in S \quad (6.17)$$

and relation (6.16),

$$(\hat{F})^*\varphi = (\hat{F}, \bar{\varphi}) = \lim_{k \rightarrow \infty} \int \hat{f}_k \varphi \, dx = \lim_{k \rightarrow \infty} \int f_k \hat{\varphi} \, dx = F^*\hat{\varphi}. \quad (6.18)$$

By (6.15) we have $(F^*)^\sim \varphi = F^*\hat{\varphi}$. Hence we obtain

$$(F^*)^\sim = (\hat{F})^*. \quad (6.19)$$

This formula says that the Fourier transform of the tempered distribution F^* is identical with the extension $(\hat{F})^*$ of the Fourier transform $\hat{F}$ of F . Hence

the definitions of $\hat{F}$ and $\tilde{F}$ are compatible if we identify every $F \in L_2(R_n)$ with its bounded extension F^* to S .

Further material on the Fourier transform, in particular a distribution-theoretical approach to the L_1 -theory and to Wiener's Tauberian theory, is contained in [9].

7. FOURIER SERIES EXPANSIONS

In this section we discuss the Fourier series expansions of the elements of the Hilbert space $L_2(W_a)$ where W_a denotes the open n -dimensional cube

$$W_a = \{x = (x_1, \dots, x_n) : -\pi a < x_i < \pi a, i = 1, \dots, n\}. \quad (7.1)$$

The second part of Lemma 2.5 shows that every function $u \in C(\bar{W}_a)$ generates a functional $F_u \in L_2(W_a)$ by formula (1.3) and that the relation (2.21) holds. In particular, the functions

$$\varphi_p(x) = (2\pi a)^{-n/2} e^{ip \cdot x/a}, \quad (7.2)$$

with $p = (p_1, \dots, p_n)$ denoting multi-indices, belong to $C^\infty(W_a) \cap L_2(W_a)$. By (2.21) we obtain

$$(\varphi_p, \varphi_q) = (2\pi a)^{-n} \int_{W_a} \exp \left[i(p - q) \cdot \frac{x}{a} \right] dx = \delta_{pq} \quad (7.3)$$

with $\delta_{pq} = 1$ if $p = q$ and $\delta_{pq} = 0$ if $p \neq q$. Hence the functions φ_p defined by (7.2) form an orthonormal system in the Hilbert space $L_2(W_a)$.

Consider, for every $F \in L_2(W_a)$, the Fourier series

$$PF = \sum_p (F, \varphi_p) \varphi_p \quad (7.4)$$

with respect to the orthonormal system (7.2). The theory of orthogonal systems in Hilbert spaces implies that the infinite sum on the right-hand side converges in $L_2(W_a)$ and that the operator P defined by (7.4) is the projection of $L_2(W_a)$ onto the smallest complete linear subspace M containing all functions φ_p . By the classical theory of trigonometric series expansions, every function of $C_0^\infty(W_a)$ can be expanded in W_a as a uniformly converging Fourier series with respect to the trigonometric system (7.2). Hence $C_0^\infty(W_a) \subset M$. Since M is complete and $C_0^\infty(W_a)$ is dense in $L_2(W_a)$, we conclude that $M = L_2(W_a)$. This shows that P is the identity operator:

$$F = \sum_p (F, \varphi_p) \varphi_p \quad \text{for every } F \in L_2(W_a). \quad (7.5)$$

Formula (7.5), together with (7.3), implies Parseval's relation

$$\|F\|^2 = \sum_p |(F, \varphi_p)|^2. \quad (7.6)$$

By (7.6) the orthonormal system (7.2) is complete. Thus, we obtain the following expansion theorem:

THEOREM 7.1. *The functions φ_p defined by (7.2) form a complete orthonormal system in the Hilbert space $L_2(W_a)$. In particular, Parseval's relation (7.6) holds for every $F \in L_2(W_a)$. Furthermore, for every $F \in L_2(W_a)$ the Fourier series (7.5) converges to F in $L_2(W_a)$.*

In the remaining part of this section we apply Theorem 7.1 to the discussion of some properties of the Sobolev spaces introduced in Section 4.

THEOREM 7.2. *For every $F \in \dot{H}_m(W_a)$ the relation*

$$\|F\|_m^2 = \sum_p \left(\sum_{|q| \leq m} a^{-2|q|} p^{2q} \right) |(F, \varphi_p)|^2 \quad (7.7)$$

holds where

$$p^{2q} = p_1^{2q_1} \cdot \dots \cdot p_n^{2q_n} \quad \text{and} \quad 0^{q_j} = 1.$$

PROOF. Since $F \in \dot{H}_m(W_a)$, Theorem 4.2 and (7.2) imply that

$$(D^q F, \varphi_p) = (-1)^{|q|} (F, D^q \varphi_p) = \left(-\frac{i}{a} \right)^{|q|} p^q (F, \varphi_p)$$

for $|q| \leq m$. By Parseval's relation (7.6), we have

$$\|D^q F\|^2 = \sum_p |(D^q F, \varphi_p)|^2 = \sum_p a^{-2|q|} p^{2q} |(F, \varphi_p)|^2.$$

This, together with (4.3), implies (7.7).

As an application of Theorem 7.2, we prove the following version of Rellich's selection theorem:

THEOREM 7.3. *Assume that Ω is a bounded open subset of R_n and consider a bounded sequence $\{F_k\}$ in the Hilbert space $\dot{H}_1(\Omega)$. Then $\{F_k\}$ contains a subsequence $\{G_k\}$ which converges in the Hilbert space $L_2(\Omega)$.*

PROOF. First we assume that $\Omega = W_a$. Then Theorem 7.2 implies that

$$\|F_k\|_1^2 = \sum_p \left(1 + \frac{p_1^2}{a^2} + \dots + \frac{p_n^2}{a^2} \right) |(F_k, \varphi_p)|^2. \quad (7.8)$$

Choose a $C > 0$ such that $\|F_k\|_1 \leq C$ for every k . By (7.8) we have also $|(F_k, \varphi_p)| \leq C$ for every k and every p . By the Bolzano-Weierstrass theorem and the usual diagonal procedure, we can select a subsequence $\{G_k\}$ of $\{F_k\}$ such that $\{(G_k, \varphi_p)\}$ converges for every p . By (7.5) we have

$$G_k = H_k^N + J_k^N \quad (7.9)$$

with

$$H_k^N = \sum_{|p| \leq N} (G_k, \varphi_p) \varphi_p, \quad J_k^N = \sum_{|p| > N} (G_k, \varphi_p) \varphi_p. \quad (7.10)$$

Since $p_1^2 + \dots + p_n^2 > N$ if $|p| > N$, we obtain by (7.8) and the choice of C

$$\begin{aligned} \|J_k^N\|^2 &= \sum_{|p| > N} |(G_k, \varphi_p)|^2 \\ &\leq \frac{1}{1 + N/a^2} \sum_{|p| > N} \left(1 + \frac{p_1^2}{a^2} + \dots + \frac{p_n^2}{a^2}\right) |(G_k, \varphi_p)|^2 \\ &\leq \frac{1}{1 + N/a^2} \|G_k\|_1^2 \leq \frac{C^2}{1 + N/a^2}. \end{aligned}$$

Hence, for every given $\epsilon > 0$ we can choose N such that $\|J_k^N\| < \epsilon/3$ for every k . Since (G_j, φ_p) converges for every p , we can choose a $N_1 > N$ such that for all $j, k > N_1$

$$\|H_j^N - H_k^N\| \leq \sum_{|p| \leq N} |(G_j, \varphi_p) - (G_k, \varphi_p)| < \frac{\epsilon}{3}$$

and hence

$$\|G_j - G_k\| \leq \|H_j^N - H_k^N\| + \|J_j^N\| + \|J_k^N\| < \epsilon.$$

This shows that $\{G_k\}$ converges in $L_2(W_a)$.

Now consider an arbitrary bounded open set $\Omega \subset R_n$ and choose a cube W_a with $\Omega \subset W_a$. Assume that the sequence $\{F_k\}$ is bounded in $\dot{H}_1(\Omega)$. Denote the norm-invariant extension of F_k to W_a by F_k^* . Lemma 4.2 implies that $\{F_k^*\}$ is bounded in $\dot{H}_1(W_a)$. Hence the argument above shows that a subsequence $\{G_k\}$ of $\{F_k\}$ exists such that the sequence $\{G_k^*\}$ of the norm-invariant extensions G_k^* of G_k converges in $L_2(W_a)$. Since $G_k - G_j$ is the restriction of $G_k^* - G_j^*$ to $C_0^\infty(\Omega)$, it follows from (2.23) that $\{G_k\}$ converges in $L_2(\Omega)$. This concludes the proof of Theorem 7.3.

8. TRANSLATIONS

First we consider the case $\Omega = R_n$. Assume that $\varphi \in C_0^\infty(R_n)$. For every $h = (h_1, \dots, h_n) \in R_n$ the translation operator T_h is defined by

$$T_h \varphi(x) = \varphi(x + h). \quad (8.1)$$

T_h is a one-to-one, linear, isometric mapping from the inner product space $C_0^\infty(R_n)$ onto itself. By the remarks in Section 6, T_h can be uniquely extended to a one-to-one, linear, isometric mapping from $L_2(R_n)$ onto $L_2(R_n)$. We show that this extension is given by

$$(T_h F) \varphi = F(T_{-h} \varphi) \quad \text{for} \quad \varphi \in C_0^\infty(R_n), \quad F \in L_2(R_n). \quad (8.2)$$

In fact, assume that $F = F_\psi$ is generated by a function $\psi \in C_0^\infty(R_n)$. Then (8.1) and (8.2) imply that

$$(T_h F_\psi) \varphi = F_\psi(T_{-h} \varphi) = \int \psi(x) \varphi(x - h) dx = \int \psi(x + h) \varphi(x) dx = F_{T_h \psi} \varphi.$$

Hence the functional $T_h F_\psi$ is generated by the function $T_h \psi$ so that the definitions (8.1) and (8.2) are compatible. Furthermore, (8.2) implies that

$$|(T_h F) \varphi| = |F(T_{-h} \varphi)| \leq \|F\| \cdot \|T_{-h} \varphi\| = \|F\| \cdot \|\varphi\|.$$

Hence, $T_h F \in L_2(R_n)$ and $\|T_h F\| \leq \|F\|$. On the other hand, since $F = T_{-h}(T_h F)$, we obtain also $\|F\| \leq \|T_h F\|$ and hence $\|T_h F\| = \|F\|$. Thus T_h is a linear, isometric mapping from $L_2(R_n)$ onto $L_2(R_n)$ which extends the mapping (8.1).

It is convenient to define also a translation operator τ_h for functionals $F \in L_2(\Omega)$ where Ω is an open subset of R_n . Consider the norm-invariant extension F^* of F to $L_2(R_n)$ introduced at the end of Section 2. Then $G = T_h F^*$ belongs to $L_2(R_n)$. Let G^r be the restriction of G to $C_0^\infty(\Omega)$ defined by (2.22). Then $F \rightarrow G^r = (T_h F^*)^r$ is a linear mapping from $L_2(\Omega)$ into $L_2(\Omega)$. We call this mapping the translation operator τ_h in $L_2(\Omega)$:

$$\tau_h F = (T_h F^*)^r \quad \text{for} \quad F \in L_2(\Omega). \quad (8.3)$$

Note that $\tau_h = T_h$ if $\Omega = R_n$. The relations (2.23), (2.26), and $\|T_h G\| = \|G\|$ for $G \in L_2(R_n)$ imply that

$$\|\tau_h F\| \leq \|F\| \quad \text{for} \quad F \in L_2(\Omega). \quad (8.4)$$

Consider a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$.

Then (2.22), (2.24), and (8.2-3) show that for every $\varphi \in C_0^\infty(\Omega)$

$$\begin{aligned} (\tau_h F) \varphi &= (T_h F^*) \varphi = F^*(T_{-h} \varphi) \\ &= \lim_{k \rightarrow \infty} \int f_k T_{-h} \varphi \, dx = \lim_{k \rightarrow \infty} \int f_k(x) \varphi(x - h) \, dx \\ &= \lim_{k \rightarrow \infty} \int f_k(x + h) \varphi(x) \, dx. \end{aligned} \quad (8.5)$$

Now assume that $F = F_u$ is generated by a function $u \in C_0(\Omega)$ and choose a sequence $\{u_k\}$ in $C_0^\infty(\Omega)$ such that $\int |u - u_k|^2 \, dx \rightarrow 0$ as $k \rightarrow \infty$. This implies, by Lemma 2.5, $\|F_u - u_k\| \rightarrow 0$ as $k \rightarrow \infty$ and hence, by (8.5) and Schwarz's inequality,

$$(\tau_h F_u) \varphi = \lim_{k \rightarrow \infty} \int u_k(x + h) \varphi(x) \, dx = \int u(x + h) \varphi(x) \, dx. \quad (8.6)$$

Thus the translate $\tau_h u = \tau_h F_u$ of a function $u \in C_0(\Omega)$ is generated by the continuous function

$$v(x) = u(x + h) \quad (x \in \Omega). \quad (8.7)$$

In particular, $v \in C(\Omega) \cap L_2(\Omega)$ for every real vector h . Note that $v = 0$ in Ω if Ω is bounded and $|h|$ is sufficiently large. Hence the mapping τ_h is, in general, not isometric and not one-to-one. We collect our results in the following theorem:

THEOREM 8.1. *The translation operator τ_h defined by (8.2) and (8.3) is a linear mapping from $L_2(\Omega)$ into $L_2(\Omega)$ with the property (8.4). Assume that $f_k \in C_0^\infty(\Omega)$ and $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. Then $\tau_h F$ is given by*

$$(\tau_h F) \varphi = \lim_{k \rightarrow \infty} \int f_k(x + h) \varphi(x) \, dx \quad \text{for} \quad \varphi \in C_0^\infty(\Omega). \quad (8.8)$$

If $F = F_u$ is generated by the function $u \in C_0(\Omega)$ then $\tau_h u = \tau_h F_u$ is generated by the restriction v of the translate $u(x + h)$ to Ω . If $\Omega = R_n$ then $\tau_h = T_h$ is a one-to-one isometric mapping from $L_2(R_n)$ onto $L_2(R_n)$.

The following two Lemmas are useful for computations with translation operators.

LEMMA 8.1. *Assume that $F, G \in L_2(\Omega)$ and $h \in R_n$. Then*

$$(F, \tau_h G) = (\tau_{-h} F, G). \quad (8.9)$$

PROOF. First we verify (8.9) for $\varphi, \psi \in C_0^\infty(\Omega)$. By Theorem 8.1 the trans-

late $\tau_{-h}\varphi$ of $\varphi \in C_0^\infty(\Omega)$ is the functional F_u generated by the continuous function $u(x) = \varphi(x - h)$ for $x \in \Omega$. Hence we obtain, by Lemma 2.3,

$$(\tau_{-h}\varphi, \psi) = (F_u, \psi) = F_u \bar{\psi} = \int \varphi(x - h) \overline{\psi(x)} dx. \quad (8.10)$$

By replacing φ by ψ and h by $-h$ in (8.10), we get

$$(\varphi, \tau_h \psi) = (\overline{\tau_h \psi}, \varphi) = \int \overline{\psi(x + h)} \varphi(x) dx = \int \overline{\psi(x)} \varphi(x - h) dx$$

and hence

$$(\varphi, \tau_h \psi) = (\tau_{-h}\varphi, \psi). \quad (8.11)$$

Now consider two functionals $F, G \in L_2(\Omega)$ and choose sequences $\{f_k\}$ and $\{g_k\}$ in $C_0^\infty(\Omega)$ such that $\|F - f_k\| \rightarrow 0$ and $\|G - g_k\| \rightarrow 0$ as $k \rightarrow \infty$. By (8.4), we have also $\|\tau_h F - \tau_h f_k\| \rightarrow 0$ and $\|\tau_h G - \tau_h g_k\| \rightarrow 0$ as $k \rightarrow \infty$. Hence (8.11) and the continuity of the inner product imply that

$$(F, \tau_h G) = \lim_{k \rightarrow \infty} (f_k, \tau_h g_k) = \lim_{k \rightarrow \infty} (\tau_{-h} f_k, g_k) = (\tau_{-h} F, G).$$

LEMMA 8.2. (a) For every $F \in L_2(R_n)$ and every differential operator D^p the relation

$$D^p(T_h F) = T_h(D^p F) \quad (8.12)$$

holds. If, in addition, $F \in H_m(R_n)$ then $T_h F \in H_m(R_n)$ and $\|T_h F\|_m = \|F\|_m$.

(b) If $\Omega \subset R_n$ and $F \in \dot{H}_m(\Omega)$ then $\tau_h F \in H_m(\Omega)$ and

$$D^p(\tau_h F) = \tau_h(D^p F). \quad (8.13)$$

Furthermore, $\|\tau_h F\|_m \leq \|F\|_m$.

PROOF. (8.12) follows from

$$\begin{aligned} [D^p(T_h F)] \varphi &= (-1)^{|p|} (T_h F) (D^p \varphi) = (-1)^{|p|} F [T_{-h}(D^p \varphi)] \\ &= (-1)^{|p|} F [D^p(T_{-h} \varphi)] = (D^p F) (T_{-h} \varphi) = [T_h(D^p F)] \varphi. \end{aligned}$$

Assume that $F \in H_m(R_n)$. Since T_h is isometric, (8.12) implies that

$$D^p(T_h F) \in L_2(R_n) \quad \text{and} \quad \|D^p(T_h F)\| = \|D^p F\|$$

if $|p| \leq m$. This concludes the proof of (a).

Now assume that $F \in \dot{H}_m(\Omega)$. By Lemma 4.2, $F^* \in H_m(R_n)$. Hence (a) implies that $T_h F^* \in H_m(R_n)$ and that

$$\|T_h F^*\|_{m, R_n} = \|F^*\|_{m, R_n} = \|F\|_{m, \Omega}.$$

Thus (b) follows from (8.3) and Lemma 4.3. In particular, (8.3), (4.9), (4.7), and (8.12) imply that

$$\begin{aligned} D^p(\tau_h F) &= D^p[(T_h F^*)^r] = [D^p(T_h F^*)]^r \\ &= [T_h(D^p F^*)]^r = [T_h(D^p F)^*]^r = \tau_h(D^p F). \end{aligned}$$

The following theorem is closely related to Lebesgue's theorem on the mean continuity of square-integrable functions:

THEOREM 8.2. *For every $F \in L_2(\Omega)$ we have*

$$\|\tau_h F - F\| \rightarrow 0 \quad \text{as} \quad h \rightarrow 0. \quad (8.14)$$

PROOF. Choose, for given $\epsilon > 0$, a function $f \in C_0^\infty(\Omega)$ such that $\|F - f\| < \epsilon$. (8.4) implies that

$$\|\tau_h F - F\| \leq \|\tau_h(F - f)\| + \|\tau_h f - f\| + \|f - F\| \leq \|\tau_h f - f\| + 2\epsilon. \quad (8.15)$$

By Theorem 8.1, the functional $\tau_h f$ is generated by the function $f(x + h)$. If $|h| < \delta = \text{dist}(\text{supp } f, \partial\Omega)$ then $f(x + h)$ belongs to $C_0^\infty(\Omega)$, and hence we have

$$\|\tau_h f - f\|^2 = \int |f(x + h) - f(x)|^2 dx. \quad (8.16)$$

Thus, we can choose $\delta_1 > 0$ such that $\|\tau_h f - f\| < \epsilon$ and hence, by (8.15), $\|\tau_h F - F\| < 3\epsilon$ if $|h| < \delta_1$. This proves (8.7).

In the remaining part of this section we discuss difference quotients

$$F_i^h = \frac{1}{h} (\tau_{he_i} F - F) \quad (8.17)$$

of functionals $F \in L_2(\Omega)$ where e_i denotes the i th unit vector and h is a real number. We show:

THEOREM 8.3. *For every $F \in \dot{H}_1(\Omega)$ we have*

$$\|F_i^h\| \leq \|F\|_1 \quad (8.18)$$

and

$$\lim_{h \rightarrow 0} \left\| F_i^h - \frac{\partial}{\partial x_i} F \right\| = 0. \quad (8.19)$$

Formula (8.19) says that the derivative $(\partial/\partial x_i)F$ of a functional $F \in \dot{H}_1(\Omega)$ can be obtained as the L_2 -limit of the corresponding difference quotient F_i^h as $h \rightarrow 0$.

PROOF OF (8.18). Choose a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ such that $\|F - f_k\|_1 \rightarrow 0$ as $k \rightarrow \infty$. By (8.8) we have

$$F_i^h \varphi = \lim_{k \rightarrow \infty} \frac{1}{h} \int [f_k(x + he_i) - f_k(x)] \varphi(x) dx. \quad (8.20)$$

By applying Schwarz's inequality twice, we obtain for $\varphi \in C_0^\infty(\Omega)$

$$\begin{aligned} & \left| \frac{1}{h} \int [f_k(x + he_i) - f_k(x)] \varphi(x) dx \right|^2 \\ & \leq \| \varphi \|^2 \frac{1}{h^2} \int |f_k(x + he_i) - f_k(x)|^2 dx \\ & = \| \varphi \|^2 \frac{1}{h^2} \int \left| \int_0^h \frac{\partial}{\partial t} f_k(x + te_i) dt \right|^2 dx \\ & \leq \| \varphi \|^2 \frac{1}{h} \int \left[\int_0^h \left| \frac{\partial}{\partial t} f_k(x + te_i) \right|^2 dt \right] dx \\ & = \| \varphi \|^2 \frac{1}{h} \int_0^h \left[\int \left| \frac{\partial}{\partial t} f_k(x + te_i) \right|^2 dx \right] dt \\ & = \| \varphi \|^2 \frac{1}{h} \int_0^h \left[\int \left| \frac{\partial}{\partial x_i} f_k(x) \right|^2 dx \right] dt \leq \| \varphi \|^2 \cdot \| f_k \|_1^2 \end{aligned}$$

and hence, by (8.20),

$$|F_i^h \varphi| \leq \| \varphi \| \overline{\lim_{k \rightarrow \infty}} \| f_k \|_1 = \| F \|_1 \cdot \| \varphi \| \quad (8.21)$$

which proves (8.18).

PROOF OF (8.19). Choose $\{f_k\}$ as above. By (8.8) we have

$$\begin{aligned} \left(F_i^h - \frac{\partial}{\partial x_i} F \right) \varphi &= \lim_{k \rightarrow \infty} \int \left[\frac{f_k(x + he_i) - f_k(x)}{h} - \frac{\partial}{\partial x_i} f_k(x) \right] \varphi(x) dx \\ &= \frac{1}{h} \lim_{k \rightarrow \infty} \int \left[\int_0^h \left(\frac{\partial}{\partial t} f_k(x + te_i) - \frac{\partial}{\partial x_i} f_k(x) \right) dt \right] \varphi(x) dx \end{aligned}$$

and hence, by Schwarz's inequality,

$$\begin{aligned} \left\| F_i^h - \frac{\partial}{\partial x_i} F \right\|^2 &\leq \frac{1}{h^2} \overline{\lim_{k \rightarrow \infty}} \int \left| \int_0^h \left(\frac{\partial}{\partial t} f_k(x + te_i) - \frac{\partial}{\partial x_i} f_k(x) \right) dt \right|^2 dx \\ &\leq \frac{1}{h} \overline{\lim_{k \rightarrow \infty}} \int \left[\int_0^h \left| \frac{\partial}{\partial t} f_k(x + te_i) - \frac{\partial}{\partial x_i} f_k(x) \right|^2 dt \right] dx \\ &= \frac{1}{h} \overline{\lim_{k \rightarrow \infty}} \int_0^h \left[\int \left| \frac{\partial}{\partial x_i} f_k(x + te_i) - \frac{\partial}{\partial x_i} f_k(x) \right|^2 dx \right] dt \\ &= \frac{1}{h} \overline{\lim_{k \rightarrow \infty}} \int_0^h \left\| T_{te_i} \left(\frac{\partial}{\partial x_i} f_k \right) - \frac{\partial}{\partial x_i} f_k \right\|^2 dt, \end{aligned}$$

where T_{te_i} denotes the translation operator in $C_0^\infty(R_n)$ defined by (8.1). Consider the extension F^* of F to $C_0^\infty(R_n)$ given by (2.24). Now Lemma 4.2 implies

$$\|F^* - f_k\|_{1, R_n} = \|F - f_k\|_{1, \Omega}. \quad (8.22)$$

Since $\|T_{te_i}G\| = \|G\|$ for every $G \in L_2(R_n)$, we obtain, by (8.22),

$$\begin{aligned} & \left\| \left[T_{te_i} \left(\frac{\partial}{\partial x_i} f_k \right) - \frac{\partial}{\partial x_i} f_k \right] - \left[T_{te_i} \left(\frac{\partial}{\partial x_i} F^* \right) - \frac{\partial}{\partial x_i} F^* \right] \right\|_{L_2(R_n)} \\ & \leq 2 \left\| \frac{\partial}{\partial x_i} f_k - \frac{\partial}{\partial x_i} F^* \right\|_{L_2(R_n)} \leq 2 \|F - f_k\|_{1, \Omega} \rightarrow 0 \quad \text{as } k \rightarrow \infty. \end{aligned}$$

Hence the integrand of the last integral converges to

$$\alpha(t) = \left\| T_{te_i} \left(\frac{\partial}{\partial x_i} F^* \right) - \frac{\partial}{\partial x_i} F^* \right\|_{L_2(R_n)}^2 \quad (8.23)$$

uniformly in t as $k \rightarrow \infty$. Hence, $\alpha(t)$ is continuous and we get, upon interchanging the order of the integration and the limit $k \rightarrow \infty$,

$$\left\| F_i^h - \frac{\partial}{\partial x_i} F \right\|^2 \leq \frac{1}{h} \int_0^h \alpha(t) dt \rightarrow 0 \quad \text{as } h \rightarrow 0. \quad (8.24)$$

Here we have also used $\alpha(0) = 0$ which is a consequence of Theorem 8.2. This completes the proof of (8.19). Now we prove the following estimate:

LEMMA 8.3. *Assume that $u \in C_b(\Omega)$ and $F \in L_2(\Omega)$. Then we have, for every real h ,*

$$\|(uF)_i^h - uF_i^h\| \leq \|F\| \cdot \sup_{x, y \in \Omega} \frac{|u(x) - u(y)|}{|x - y|}. \quad (8.25)$$

PROOF. Choose a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. By Lemma 2.3, Lemma 8.1, and Lemma 5.3 we obtain for every $\varphi \in C_0^\infty(\Omega)$ and for every $h \in R_n$

$$\begin{aligned} [\tau_h(uF)]\varphi &= (\tau_h(uF), \bar{\varphi}) = (uF, \tau_{-h}\bar{\varphi}) \\ &= (F, \bar{u}\tau_{-h}\bar{\varphi}) = \lim_{k \rightarrow \infty} (f_k, \bar{u}\tau_{-h}\bar{\varphi}). \end{aligned} \quad (8.26)$$

Theorem 8.1 implies that $\tau_{-h}\bar{\varphi}$ belongs to $C_b(\Omega) \cap L_2(\Omega)$ and is generated by the function $u(x) = \overline{\varphi(x - h)}$ if $x \in \Omega$. Hence, by Lemma 5.2 the functional $\bar{u}\tau_{-h}\bar{\varphi}$ is generated by the function $u(x) \cdot \overline{\varphi(x - h)}$. Thus Lemma 2.3 yields

$$(f_k, \bar{u}\tau_{-h}\bar{\varphi}) = \overline{(\bar{u}\tau_{-h}\bar{\varphi}, f_k)} = \int u(x) \varphi(x - h) f_k(x) dx$$

and hence, by (8.26),

$$[\tau_h(uF)]\varphi = \lim_{k \rightarrow \infty} \int u(x+h)\varphi(x)f_k(x+h)dx. \quad (8.27)$$

Since $\|u(\tau_h F) - u(\tau_h f_k)\| \rightarrow 0$ as $k \rightarrow \infty$ by (5.3) and (8.4), we obtain in a similar way

$$\begin{aligned} [u(\tau_h F)]\varphi &= (u \cdot \tau_h F, \bar{\varphi}) \\ &= \lim_{k \rightarrow \infty} (u \cdot \tau_h f_k, \bar{\varphi}) = \lim_{k \rightarrow \infty} \int u(x)f_k(x+h)\varphi(x)dx. \end{aligned} \quad (8.28)$$

(8.27) and (8.28) imply that for real h and $\varphi \in C_0^\infty(\Omega)$

$$\begin{aligned} |[(uF)_i^h - uF_i^h]\varphi| &= \left| \frac{1}{h} [\tau_{he_i}(uF) - u(\tau_{he_i}F)]\varphi \right| \\ &= \lim_{k \rightarrow \infty} \left| \int \frac{u(x+he_i) - u(x)}{h} f_k(x+he_i)\varphi(x)dx \right| \\ &\leq \sup_{x,y \in \Omega} \frac{|u(x) - u(y)|}{|x - y|} \cdot \|\varphi\| \cdot \overline{\lim_{k \rightarrow \infty}} \|f_k\| \\ &= \|F\| \cdot \|\varphi\| \cdot \sup_{x,y \in \Omega} \frac{|u(x) - u(y)|}{|x - y|} \end{aligned}$$

which proves (8.25).

We conclude this section with a proof of the following theorem which is useful for the discussion of differentiability properties of weak solutions of partial differential equations:

THEOREM 8.4. *Assume that $F \in \dot{H}_j(\Omega)$ with $j \geq 1$ and that constants $K > 0$ and $\delta > 0$ exist such that $\|F_i^h\|_j \leq K$ for all real h with $|h| \leq \delta$. Then $(\partial/\partial x_i)F \in H_j(\Omega)$.*

PROOF. Since $\|D^p F_i^h\|$ is bounded as $h \rightarrow 0$ for every differential operator D^p of order $|p| \leq j$, we can select, by the weak compactness theorem for Hilbert spaces, a sequence $\{h_k\}$ with $h_k \rightarrow 0$ such that $\{D^p F_i^{h_k}\}$ converges weakly to an element $F_{i,p} \in L_2(\Omega)$ for every $p = (p_1, \dots, p_n)$ with $|p| \leq j$:

$$(D^p F_i^{h_k}, G) \rightarrow (F_{i,p}, G) \quad \text{as } k \rightarrow \infty \quad (8.29)$$

for every $G \in L_2(\Omega)$. Set $F_{i,0} = F_{i,(0,\dots,0)} = J_i$. By Lemma 2.3 and Theorem 4.2, we obtain for $\varphi \in C_0^\infty(\Omega)$ and $|p| \leq j$

$$\begin{aligned} F_{i,p}\varphi &= (F_{i,p}, \bar{\varphi}) = \lim_{k \rightarrow \infty} (D^p F_i^{h_k}, \bar{\varphi}) \\ &= (-1)^{|p|} \lim_{k \rightarrow \infty} (F_i^{h_k}, D^p \bar{\varphi}) = (-1)^{|p|} (J_i, D^p \bar{\varphi}) \\ &= (-1)^{|p|} J_i(D^p \varphi) = (D^p J_i)\varphi \end{aligned}$$

and hence $D^p J_i = F_{i,p} \in L_2(\Omega)$. This yields $J_i \in H_j(\Omega)$. On the other hand, (8.19) implies that $\|F_i^{h_k} - (\partial/\partial x_i) F\| \rightarrow 0$ as $k \rightarrow \infty$. Thus we get

$$\left(\frac{\partial}{\partial x_i} F, G\right) = \lim_{k \rightarrow \infty} (F_i^{h_k}, G) = (J_i, G) \quad (8.30)$$

for every $G \in L_2(\Omega)$ and hence $(\partial/\partial x_i) F = J_i \in H_j(\Omega)$. This completes the proof of Theorem 8.4.

9. MOLLIFIERS

In the following two sections we discuss the approximation of functionals $F \in H_m(\Omega)$ by C^∞ -functions with respect to the norm (4.3). Let h be a function with the properties (2.13). If $u \in C_0^m(R_n)$, then the proofs of Lemma 2.5 and Lemma 4.1 show that the function

$$(J_\epsilon u)(x) = \frac{1}{\epsilon^n} \int u(y) h\left(\frac{x-y}{\epsilon}\right) dy \quad (\epsilon > 0) \quad (9.1)$$

belongs to $C_0^\infty(R_n)$ and that $\|J_\epsilon u - u\|_m \rightarrow 0$ as $\epsilon \rightarrow 0$. The operator J_ϵ , mapping $C_0(R_n)$ into $C_0^\infty(R_n)$, is called a mollifier. Set

$$i_\epsilon(x) = \frac{1}{\epsilon^n} h\left(\frac{x}{\epsilon}\right). \quad (9.2)$$

Then we have $J_\epsilon u = j_\epsilon * u$ where the star denotes the convolution product. In order to extend J_ϵ to $L_2(R_n)$, we write (9.1) in the form

$$(J_\epsilon u)(x) = (F_u)_y j_\epsilon(x-y) \quad (9.3)$$

where F_u is defined by (1.3) and the subscript y indicates that the functional F_u is applied to a function of y . This suggests defining J_ϵ for $F \in L_2(R_n)$ by

$$(J_\epsilon F)(x) = F_y j_\epsilon(x-y). \quad (9.4)$$

Note that $j_\epsilon(x-y)$, as a function of y , belongs to $C_0^\infty(R_n)$. In terms of distribution theory, $J_\epsilon F$ is the convolution product of the functional F and the C_0^∞ -function j_ϵ . We collect the main properties of the operator J_ϵ in the following theorem:

THEOREM 9.1. *Assume that $F \in L_2(R_n)$ and define $J_\epsilon F$ by (9.4). Then $J_\epsilon F \in C^\infty(R_n) \cap L_2(R_n)$. Furthermore,*

$$\|J_\epsilon F\| \leq \|F\| \quad (9.5)$$

and

$$\|F - J_\epsilon F\| \rightarrow 0 \quad \text{as} \quad \epsilon \rightarrow 0. \quad (9.6)$$

If, in addition, $F \in H_m(R_n)$ then also $J_\epsilon F \in H_m(R_n)$ and

$$D^p(J_\epsilon F) = J_\epsilon(D^p F) \quad \text{if} \quad |p| \leq m. \quad (9.7)$$

Furthermore,

$$\|F - J_\epsilon F\|_m \rightarrow 0 \quad \text{as} \quad \epsilon \rightarrow 0. \quad (9.8)$$

PROOF. By (9.4) we have, for every $x \in R_n$ and every $\epsilon > 0$,

$$|(J_\epsilon F)(z) - (J_\epsilon F)(x)| \leq \|F\| \cdot \left[\int |j_\epsilon(z-y) - j_\epsilon(x-y)|^2 dy \right]^{1/2} \rightarrow 0 \quad (9.9)$$

as $z \rightarrow x$ and

$$\begin{aligned} & \left| \frac{(J_\epsilon F)(x + he_i) - (J_\epsilon F)(x)}{h} - F_y \left(\frac{\partial}{\partial x_i} j_\epsilon(x-y) \right) \right| \\ & \leq \|F\| \cdot \left[\int \left| \frac{j_\epsilon(x + he_i - y) - j_\epsilon(x-y)}{h} - \frac{\partial}{\partial x_i} j_\epsilon(x-y) \right|^2 dy \right]^{1/2} \rightarrow 0 \end{aligned} \quad (9.10)$$

as $h \rightarrow 0$. Hence $J_\epsilon F$ is continuous and has first order derivatives given by

$$\frac{\partial}{\partial x_i} (J_\epsilon F)(x) = F_y \left(\frac{\partial}{\partial x_i} j_\epsilon(x-y) \right).$$

By repeating this argument we conclude that $J_\epsilon F$ has continuous derivatives of arbitrary order which are given by

$$D^p(J_\epsilon F)(x) = F_y(D_x^p j_\epsilon(x-y)). \quad (9.11)$$

This shows that $J_\epsilon F \in C^\infty(R_n)$. Now choose a sequence $\{f_k\}$ in $C_0^\infty(R_n)$ such that $\|F - f_k\| \rightarrow 0$. For every $\varphi \in C_0^\infty(R_n)$ we have

$$\begin{aligned} \left| \int (J_\epsilon F) \cdot \varphi dx \right| &= \left| \int \varphi(x) F_y(j_\epsilon(x-y)) dx \right| \\ &= \left| \int \varphi(x) \left[\lim_{k \rightarrow \infty} \int f_k(y) j_\epsilon(x-y) dy \right] dx \right|. \end{aligned} \quad (9.12)$$

Since

$$\left| \int [f_k(y) - f_j(y)] j_\epsilon(x-y) dy \right|^2 \leq \|f_k - f_j\|^2 \int j_\epsilon(y)^2 dy,$$

the integrand of the x -integration in (9.12) converges uniformly in x as $k \rightarrow \infty$. This implies, since the domain of the x -integration is contained in

the bounded set $\text{supp } \varphi$, that the order of the limit $k \rightarrow \infty$ and the integration with respect to x can be interchanged. Hence we obtain

$$\begin{aligned} \left| \int (J_\epsilon F) \cdot \varphi \, dx \right|^2 &= \left| \lim_{k \rightarrow \infty} \int \varphi(x) \left[\int f_k(y) j_\epsilon(x-y) \, dy \right] dx \right|^2 \\ &\leq \| \varphi \|^2 \overline{\lim_{k \rightarrow \infty}} \int \left| \int f_k(y) j_\epsilon(x-y) \, dy \right|^2 dx. \end{aligned} \quad (9.13)$$

By (9.2) and (2.13), we have

$$\int j_\epsilon(x-y) \, dy = \int j_\epsilon(x-y) \, dx = \int j_\epsilon(x) \, dx = 1$$

and hence, by Schwarz's inequality,

$$\begin{aligned} &\int \left| \int f_k(y) j_\epsilon(x-y) \, dy \right|^2 dx \\ &= \int \left| \int f_k(y) [j_\epsilon(x-y)]^{1/2} \cdot [j_\epsilon(x-y)]^{1/2} \, dy \right|^2 dx \\ &\leq \int \left[\int |f_k(y)|^2 j_\epsilon(x-y) \, dy \right] \left[\int j_\epsilon(x-y) \, dy \right] dx \\ &= \int |f_k(y)|^2 \left[\int j_\epsilon(x-y) \, dx \right] dy = \int |f_k(y)|^2 dy = \|f_k\|^2. \end{aligned}$$

By inserting this estimate in (9.13), we obtain

$$\left| \int (J_\epsilon F) \cdot \varphi \, dx \right| \leq \| \varphi \| \cdot \overline{\lim_{k \rightarrow \infty}} \|f_k\| = \|F\| \cdot \| \varphi \|. \quad (9.14)$$

This inequality shows that the functional generated by $J_\epsilon F$ belongs to $L_2(R_n)$. Thus we have proved that $J_\epsilon F$ belongs to $C^\infty(R_n) \cap L_2(R_n)$ in the sense explained in Section 2 (after the proof of Lemma 2.4) and that the estimate (9.5) holds.

In order to verify (9.6), choose for any $\eta > 0$ a function $f \in C_0^\infty(R_n)$ such that $\|F - f\| < \eta$. By (9.5), we have

$$\|F - J_\epsilon F\| \leq \|F - f\| + \|f - J_\epsilon f\| + \|J_\epsilon(f - F)\| \leq \|f - J_\epsilon f\| + 2\eta. \quad (9.15)$$

Since $J_\epsilon f$ is given by (9.1), it follows from the argument used in the proof of Lemma 2.5 that $\|J_\epsilon f - f\| \rightarrow 0$ as $\epsilon \rightarrow 0$. Hence (9.15) implies that $\|F - J_\epsilon F\| < 3\eta$ if ϵ is sufficiently small. This proves (9.6).

Now assume that $F \in H_m(R_n)$. By (9.11) and (3.1) we obtain for every $x \in R_n$

$$\begin{aligned} D^p(J_\epsilon F)(x) &= F_y(D_x p j_\epsilon(x-y)) = (-1)^{|p|} F_y(D_y p j_\epsilon(x-y)) \\ &= (D^p F)_y j_\epsilon(x-y) = (J_\epsilon(D^p F))(x) \end{aligned}$$

which proves (9.7). Since $D^p F \in L_2(R_n)$ if $|p| \leq m$, the first part of Theorem 9.1 shows that $D^p(J_\epsilon F) = J_\epsilon(D^p F)$ belongs to $L_2(R_n)$ and that

$$\|D^p F - D^p(J_\epsilon F)\| = \|D^p F - J_\epsilon(D^p F)\| \rightarrow 0$$

as $\epsilon \rightarrow 0$. Hence we have $J_\epsilon F \in H_m(R_n)$ and $\|F - J_\epsilon F\|_m \rightarrow 0$ as $\epsilon \rightarrow 0$. This completes the proof of Theorem 9.1.

Theorem 9.1 says that every functional $F \in H_m(R_n)$ can be approximated by a sequence of C^∞ -functions with respect to the m -norm defined in (4.3). Theorem 9.1 can be used to derive the following theorem on approximation in the space $H_m(\Omega)$:

THEOREM 9.2. *Assume that $F \in H_m(\Omega)$ and $\psi \in C_0^\infty(\Omega)$. Then $\psi F \in \dot{H}_m(\Omega)$.*

PROOF. Note that $\varphi\psi \in C_0^\infty(\Omega)$ for every $\varphi \in C_0^\infty(R_n)$. Hence the formula

$$G\varphi = F(\varphi \cdot \psi) \quad \text{for} \quad \varphi \in C_0^\infty(R_n) \quad (9.16)$$

defines a linear functional G on $C_0^\infty(R_n)$. We show that $G \in H_m(R_n)$. It follows by induction that for every multi-index p

$$(D^p G)\varphi = \sum_{|q|+|r|=|p|} c_{qr}^{(p)}(D^q F)(\varphi D^r \psi) \quad (9.17)$$

with suitable non-negative integers $c_{qr}^{(p)}$ depending on the multi-indices p, q, r . In fact, (9.16) is a representation of the form (9.17) for $p = (0, \dots, 0)$. Assume that (9.17) holds for a given p . Then the formulas

$$\left(\frac{\partial}{\partial x_i} D^p G\right)\varphi = - \sum_{|q|+|r|=|p|} c_{qr}^{(p)}(D^q F) \left(\frac{\partial \varphi}{\partial x_i} D^r \psi\right)$$

and

$$\begin{aligned} (D^q F) \left(\frac{\partial \varphi}{\partial x_i} D^r \psi\right) &= (D^q F) \left(\frac{\partial}{\partial x_i} (\varphi D^r \psi) - \varphi \frac{\partial}{\partial x_i} D^r \psi\right) \\ &= - \left(\frac{\partial}{\partial x_i} D^q F\right) (\varphi D^r \psi) - (D^q F) \left(\varphi \frac{\partial}{\partial x_i} D^r \psi\right) \end{aligned}$$

yield a similar representation for $(\partial/\partial x_i) D^p G$. This concludes the proof of (9.17). If $|p| \leq m$, then (9.17) implies that

$$\|(D^p G)\varphi\| \leq \sum_{|q|+|r|=|p|} c_{qr}^{(p)} \text{Max} |D^r \psi(x)| \cdot \|F\|_m \cdot \|\varphi\|$$

for every $\varphi \in C_0^\infty(R_n)$. Thus we obtain $D^p G \in L_2(R_n)$ if $|p| \leq m$ and hence $G \in H_m(R_n)$. By Theorem 9.1 we have $J_\epsilon G \in C^\infty(R_n) \cap H_m(R_n)$ and

$$\|G - J_\epsilon G\|_m \rightarrow 0 \quad \text{as} \quad \epsilon \rightarrow 0. \quad (9.18)$$

We show that $J_\epsilon G \in C_0^\infty(\Omega)$ if $\epsilon < \delta = \text{dist}(\text{supp } \psi, \partial\Omega)$. In fact, since $j_\epsilon(x) = 0$ if $|x| \geq \epsilon$, we have

$$(J_\epsilon G)(x) = G_\nu j_\epsilon(x - y) = F_\nu(\psi(y) j_\epsilon(x - y)) = 0 \quad (9.19)$$

if $\epsilon < \text{dist}(x, \text{supp } \psi)$. Hence $\epsilon < \delta$ implies that $(J_\epsilon G)(x) = 0$ for every $x \in \Omega$ with $\text{dist}(x, \partial\Omega) < \delta - \epsilon$. Thus we have proved:

LEMMA 9.1. *Assume that $F \in H_m(\Omega)$ and $\psi \in C_0^\infty(\Omega)$. Let G be defined by (9.16). Then $G \in H_m(R_n)$ and $J_\epsilon G \in C_0^\infty(\Omega)$ if $\epsilon < \delta = \text{dist}(\text{supp } \psi, \partial\Omega)$. Furthermore, $\|G - J_\epsilon G\|_m \rightarrow 0$ in $H_m(R_n)$ as $\epsilon \rightarrow 0$.*

Note that ψF is the restriction of G to $C_0^\infty(\Omega)$. Hence Lemma 2.3 implies that $\|\psi F - J_\epsilon G\|_m \rightarrow 0$ as $\epsilon \rightarrow 0$. Since $J_\epsilon G \in C_0^\infty(\Omega)$ if $\epsilon < \delta$, we conclude that $\psi F \in \dot{H}_m(\Omega)$. This completes the proof of Theorem 9.2.

We conclude this section with the following generalization of Theorem 9.2:

THEOREM 9.3. *Assume that $F \in H_m(\Omega)$ and that $\psi \in C_0^\infty(\Omega)$. Then $\tau_h(\psi F) \in \dot{H}_m(\Omega)$ if $|h| < \delta = \text{dist}(\text{supp } \psi, \partial\Omega)$.*

PROOF. Let G be defined by (9.16). Then G coincides with the norm-invariant extension $(\psi F)^*$ of ψF to $L_2(R_n)$. In fact, choose a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. Then, by (5.3), $\|\psi F - \psi f_k\| \rightarrow 0$ and hence, by (2.24),

$$(\psi F)^* \varphi = \lim_{k \rightarrow \infty} \int \psi f_k \varphi \, dx = F(\varphi \psi) = G\varphi \quad (9.20)$$

for every $\varphi \in C_0^\infty(R_n)$. Since $(\psi F)^* = G$, (8.2-3) implies that

$$[\tau_h(\psi F)] \varphi = (\psi F)^* (T_{-h} \varphi) = G(T_{-h} \varphi) = (T_h G) \varphi$$

for $\varphi \in C_0^\infty(\Omega)$. Hence $\tau_h(\psi F)$ is the restriction of $T_h G$ to $C_0^\infty(\Omega)$. By Lemma 9.1 we have $G \in H_m(R_n)$ and hence, by Lemma 8.2(a), $T_h G \in H_m(R_n)$. Thus Theorem 9.1 yields

$$\|T_h G - J_\epsilon(T_h G)\|_m \rightarrow 0 \quad \text{as} \quad \epsilon \rightarrow 0. \quad (9.21)$$

On the other hand, (9.19) shows that

$$[J_\epsilon(T_h G)](x) = (T_h G)_\nu j_\epsilon(x - y) = G_\nu j_\epsilon(x + h - y) = 0$$

if $\epsilon < \text{dist}(x + h, \text{supp } \psi)$. This condition is satisfied if $|h| < \delta$, $\epsilon < \delta - |h|$, and $\text{dist}(x, R_n - \Omega) < \delta - |h| - \epsilon$. Thus $J_\epsilon(T_h G) \in C_0^\infty(\Omega)$ if $|h| < \delta$ and $\epsilon < \delta - |h|$. Since $\tau_h(\psi F)$ is the restriction of $T_h G$, we conclude from (9.21) and Lemma 4.3 that $\|\tau_h(\psi F) - J_\epsilon(T_h G)\|_m \rightarrow 0$ as $\epsilon \rightarrow 0$. This completes the proof of Theorem 9.3.

Finally we remark that it is easy to show that a functional $F_0 \in L_2(\Omega)$ can be represented in the form $F_0 = \psi F$ with $\psi \in C_0^\infty(\Omega)$ and $F \in L_2(\Omega)$ if and only if F has compact support in the sense of distribution theory (compare, for example, [2], Definition 1.3.2). Hence Theorems 9.2 and 9.3 can be formulated as statements on functionals $F \in H_m(\Omega)$ with compact support.

10. APPROXIMATION BY C^∞ -FUNCTIONS

In the last section we have shown that a functional $F \in H_m(\Omega)$ can be approximated in the norm (4.3) by a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ under the additional assumption that the support of F is compact. In this section we discuss approximations of arbitrary functionals $F \in H_m(\Omega)$ by sequences in $C^\infty(\Omega)$. N. G. Meyers and J. Serrin [4] have proved the following result for an arbitrary open set $\Omega \subseteq R_n$:

THEOREM 10.1. *Assume that $F \in H_m(\Omega)$. Then there exists a sequence $\{f_k\}$ of functions $f_k \in C^\infty(\Omega) \cap H_m(\Omega)$ such that*

$$\|F - f_k\|_m \rightarrow 0 \quad \text{as} \quad k \rightarrow \infty. \quad (10.1)$$

In the first part of this section we reformulate the original proof in a way which avoids Lebesgue integration and is entirely based on the concepts of this paper. In the second part we discuss the question under which conditions the sequence $\{f_k\}$ in Theorem 10.1 can be chosen such that each f_k can be extended to $C_0^\infty(R_n)$. We shall prove a result (Theorem 10.2) which is related to [1], Theorem 2.1 and yields a sufficient condition.

PROOF OF THEOREM 10.1. Consider the open sets

$$\Omega_j = \left\{ x : x \in \Omega, |x| < j, \text{dist}(x, R_n - \Omega) > \frac{1}{j} \right\} \quad (10.2)$$

and set

$$A_1 = \Omega_2, \quad A_j = \Omega_{j+1} - \overline{\Omega_{j-1}} \quad \text{if} \quad j \geq 2. \quad (10.3)$$

We show that a sequence $\{\psi_j\}$ exists such that $\psi_j \in C_0^\infty(A_j)$ and

$$\sum_{j=1}^{\infty} \psi_j = 1 \quad \text{in} \quad \Omega. \quad (10.4)$$

Note that $\delta_j = \text{dist}(\Omega_j, \partial\Omega_{j+1}) > 0$ since $\bar{\Omega}_j$ is a compact subset of Ω_{j+1} . Set

$$K_1 = \left\{ x \in A_1 = \Omega_2 : \text{dist}(x, \partial\Omega_2) \geq \frac{\delta_1}{2} \right\},$$

$$K_j = \left\{ x \in A_j : \text{dist}(x, \partial\Omega_{j+1}) \geq \frac{\delta_j}{2} \text{ and } \text{dist}(x, \partial\Omega_{j-1}) \geq \frac{\delta_{j-1}}{2} \right\} \quad \text{if } j \geq 2.$$

It is easy to verify that $\Omega = \cup K_j$. Since K_j is a compact subset of the open set A_j , there exists a function $\xi_j \in C_0^\infty(A_j)$ such that $0 \leq \xi_j \leq 1$ and $\xi_j(x) = 1$ if $x \in K_j$ (apply, for example, [2], Theorem 1.2.2). By (10.2-3), every $x \in \Omega$ lies in at most two sets A_j and hence has an open neighborhood $U(x)$ such that at most two terms of the series

$$\xi = \sum_{j=1}^{\infty} \xi_j \quad (10.5)$$

are different from zero in $U(x)$. This shows that $\xi \in C^\infty(\Omega)$. Since $\xi_j \geq 0$ and $\xi_j(x) = 1$ if $x \in K_j$, ξ has no zeros in Ω . Hence the functions $\psi_j = \xi_j/\xi$ have the required properties.

Assume that $F \in H_m(\Omega)$ and set

$$G_j \varphi = F(\varphi \psi_j) \quad \text{for} \quad \varphi \in C_0^\infty(R_n). \quad (10.6)$$

Since $\psi_j \in C_0^\infty(A_j)$, Lemma 9.1 implies that for any given $\epsilon > 0$ a function $\chi_j \in C_0^\infty(A_j)$ exists such that

$$\|G_j - \chi_j\|_m < \epsilon \cdot 2^{-j} \quad (10.7)$$

in $H_m(R_n)$. Set

$$v = \sum_{j=1}^{\infty} \chi_j. \quad (10.8)$$

We conclude as above that $v \in C^\infty(\Omega)$. Consider a fixed $\varphi \in C_0^\infty(\Omega)$ and choose a $k > 0$ such that $\text{supp } \varphi \subset \Omega_k$. (10.3) implies that $\text{supp } \psi_j \subset A_j \subset R_n - \Omega_k$ if $j \geq k+1$. Hence $\psi_j D^p \varphi$ vanishes if $j \geq k+1$. Thus we obtain, by (10.4) and (10.6),

$$\begin{aligned} (D^p F) \varphi &= (-1)^{|p|} F(D^p \varphi) = (-1)^{|p|} F\left(\sum_{j=1}^k \psi_j D^p \varphi\right) \\ &= (-1)^{|p|} \sum_{j=1}^k G_j(D^p \varphi) = \sum_{j=1}^k (D^p G_j) \varphi. \end{aligned}$$

Since $\text{supp } \chi_j \subset A_j$, we have $\varphi D^p \chi_j = 0$ for $j \geq k+1$ and hence, by (10.8),

$$\varphi D^p v = \varphi \sum_{j=1}^k D^p \chi_j.$$

Thus we conclude that for $|p| \leq m$

$$\begin{aligned} \left| (D^p F) \varphi - \int \varphi D^p v \, dx \right| &= \left| \sum_{j=1}^k \left[(D^p G_j) \varphi - \int \varphi D^p \chi_j \, dx \right] \right| \\ &\leq \|\varphi\| \sum_{j=1}^k \|G_j - \chi_j\|_m \leq \epsilon \|\varphi\| \sum_{j=1}^k 2^{-j} \leq \epsilon \|\varphi\|. \end{aligned}$$

Since this estimate holds for every $\varphi \in C_0^\infty(\Omega)$, we obtain

$$\|D^p F - D^p v\| \leq \epsilon \quad \text{for} \quad |p| \leq m \quad (10.9)$$

and hence

$$\|F - v\|_m \leq c(m) \epsilon \quad (10.10)$$

where $c(m)$ depends only on m . In particular, we have $v \in C^\infty(\Omega) \cap H_m(\Omega)$. The estimate (10.10) shows that we can find a sequence $\{f_k\}$ with the required properties. This concludes the proof of Theorem 10.1.

Note that the representation (10.8) does not imply that the approximating function v has a bounded or even continuous extension to $\bar{\Omega}$ since the number of terms χ_j having non-zero values in a given neighborhood of a boundary point may not be finite. Thus the above argument does not indicate that the approximation in Theorem 10.1 can be achieved by functions $f_k \in C_0^\infty(R_n)$. It seems that approximations with this additional property have only been established under certain restrictions for Ω . We shall use the following property:

DEFINITION 10.1. An open set $\Omega \subseteq R_n$ has the segment property if for every $x_0 \in \partial\Omega$ an (open) neighborhood $O(x_0)$ of x_0 and a vector $y(x_0) \in R_n$ exist such that $x + ty(x_0) \in \Omega$ if $0 < t < 1$ and $x \in \bar{\Omega} \cap O(x_0)$.

We prove:

THEOREM 10.2. Assume that Ω has the segment property and that $F \in H_m(\Omega')$ for every bounded open subset Ω' of Ω (in the sense stated at the end of Section 4). Then there exists for every bounded open subset Ω_0 of Ω a sequence $\{f_k\}$ in $C_0^\infty(R_n)$ such that $\|F - f_k\|_m \rightarrow 0$ in $H_m(\Omega_0)$.

Note that, by Lemma 4.3, the second assumption of Theorem 10.2 is satisfied if $F \in H_m(\Omega)$. Theorem 10.2 is a variant of [1], Theorem 2.1. Our

proof follows the argument in [1] with some modifications required by our interpretation of the elements of $L_2(\Omega)$ as linear functionals.

PROOF OF THEOREM 10.2. Since $\partial\Omega \cap \partial\Omega_0$ is compact, we can choose a finite number of points $x_1, \dots, x_N \in \partial\Omega$ such that

$$\partial\Omega \cap \partial\Omega_0 \subset \bigcup_{i=1}^N O_i \quad (10.11)$$

where $O_i = O(x_i)$ is the open set associated to x_i by the segment property. Set

$$K = \bar{\Omega}_0 \cap \left(R_n - \bigcup_{i=1}^N O_i \right). \quad (10.12)$$

Since K is compact and $K \subset \Omega$, we have $\text{dist}(K, \partial\Omega) > 0$. Hence there exists an open set O_0 such that $K \subset O_0 \subset \subset \Omega$.² By (10.12), we have $\bar{\Omega}_0 \subset O_0 \cup \dots \cup O_N$. Now choose open sets $O'_0, \dots, O'_N$ such that $O'_i \subset \subset O_i$ and $\bar{\Omega}_0 \subset O'_0 \cup \dots \cup O'_N$ and functions $\xi_0, \dots, \xi_N$ and that $\xi_i \in C_0^\infty(O'_i)$ and

$$\sum_{j=0}^N \xi_j = 1 \quad \text{in} \quad \Omega_0 \quad (10.13)$$

(for example, by applying [1], Theorem 1.9). Set $F_i = \xi_i F$. Since $F = F_0 + \dots + F_N$ in Ω_0 , it is sufficient to show that each functional F_i can be approximated in Ω_0 by a sequence $\{f_k\}$ in the sense of Theorem 10.2. First we consider F_0 . Since $\Omega_0 \cup O_0$ is a bounded open subset of Ω , we have $F \in H_m(\Omega_0 \cup O_0)$, by the second assumption of Theorem 10.2. Note that $\xi_0 \in C_0^\infty(\Omega_0 \cup O_0)$ since $O'_0 \subset \Omega_0 \cup O_0$. Hence Theorem 9.2 yields $F_0 = \xi_0 F \in \dot{H}_m(\Omega_0 \cup O_0)$. Thus, we can choose a sequence $\{f_k^0\}$ in $C_0^\infty(\Omega_0 \cup O_0) \subset C_0^\infty(R_n)$ such that $\|F_0 - f_k^0\|_{m, \Omega_0 \cup O_0} \rightarrow 0$ as $k \rightarrow \infty$. By Lemma 4.3 we have $\|F_0 - f_k^0\|_{m, \Omega_0} \leq \|F_0 - f_k^0\|_{m, \Omega_0 \cup O_0}$ and hence $\|F_0 - f_k^0\|_m \rightarrow 0$ as $k \rightarrow \infty$ in Ω_0 . This establishes the required approximation for F_0 .

Next we discuss $F_1 = \xi_1 F$. Set $\Omega_1 = \Omega_0 \cup (\Omega \cap O_1')$. By the second assumption of Theorem 10.2, F belongs to $H_m(\Omega_1)$. Choose a sequence $\{f_k\}$ in $C_0^\infty(\Omega_1)$ such that $\|F - f_k\|_{L_2(\Omega_1)} \rightarrow 0$ as $k \rightarrow \infty$. Theorem 5.1 implies that $F_1 = \xi_1 F \in H_m(\Omega_1)$ and that $\|F_1 - \xi_1 f_k\|_{L_2(\Omega_1)} \rightarrow 0$ as $k \rightarrow \infty$. Hence, by (2.24), the norminvariant extension F_1^* of F_1 to $L_2(R_n)$ is given by

$$F_1^* \varphi = \lim_{k \rightarrow \infty} \int \xi_1 f_k \varphi \, dx \quad \text{for} \quad \varphi \in C_0^\infty(R_n). \quad (10.14)$$

² See Section 0 for the notation $\subset \subset$.

We show that $F_1^* \in H_m(R_n - \Gamma)$ where $\Gamma = \bar{O}_1' \cap \partial\Omega$. Consider a fixed $\varphi \in C_0^\infty(R_n - \Gamma)$. Note that $\text{dist}(\text{supp } \varphi \cap \text{supp } \xi_1, \partial\Omega) > 0$. Hence $K_1 = \text{supp } \varphi \cap \text{supp } \xi_1 \cap \bar{\Omega}$ is a compact subset of $\Omega \cap O_1'$. Choose a $\zeta \in C_0^\infty(\Omega \cap O_1')$ such that $\zeta = 1$ in K_1 . Since $\text{supp}(\varphi \cdot \xi_1 \cdot f_k) \subseteq K_1$, we have $\xi_1 f_k D^p \varphi = \xi_1 f_k D^p(\zeta \varphi)$. Thus, (10.14) implies that for $|p| \leq m$

$$\begin{aligned} (D^p F_1^*) \varphi &= (-1)^{|p|} F_1^*(D^p \varphi) = (-1)^{|p|} \lim_{k \rightarrow \infty} \int \xi_1 f_k D^p \varphi \, dx \\ &= (-1)^{|p|} \lim_{k \rightarrow \infty} \int \xi_1 f_k D^p(\zeta \varphi) \, dx = (-1)^{|p|} F_1(D^p(\zeta \varphi)) \\ &= (D^p F_1)(\zeta \varphi) \end{aligned}$$

and hence, since $\text{supp } \zeta \varphi \in \Omega \cap O_1' \subset \Omega_1$ and $F_1 \in H_m(\Omega_1)$,

$$|(D^p F_1^*) \varphi| \leq \|D^p F_1\| \cdot \|\varphi\| \cdot \text{Sup } |\zeta(x)|.$$

Since this estimate holds for every $\varphi \in C_0^\infty(R_n - \Gamma)$ and $|p| \leq m$, we conclude that $F_1^* \in H_m(R_n - \Gamma)$.

Now we consider the set

$$\Gamma_h = \Gamma - hy_1 \quad \text{with} \quad \Gamma = \bar{O}_1' \cap \partial\Omega \quad (10.15)$$

where $y_1 = y(x_1)$ is the vector associated to $x_1 \in \partial\Omega$ by the segment property. We show that

$$\text{dist}(\Gamma_h, \Omega) > 0 \quad \text{if} \quad 0 < h < h_0 = \text{Min}(1, \delta |y_1|^{-1}) \quad (10.16)$$

where $\delta = \text{dist}(O_1', \partial O_1)$. In fact, if $z \in \Gamma_h$ then

$$\text{dist}(z, O_1') \leq \text{dist}(z, \Gamma) \leq h |y_1| < \delta$$

and hence $z \in O_1$. This shows that $\Gamma_h \subset O_1$. Thus $z \in \Gamma_h \cap \bar{\Omega}$ implies $z \in O_1 \cap \bar{\Omega}$ and hence, by the segment property, $z + hy_1 \in \Omega$. On the other hand, since $z \in \Gamma_h$, we have $z + hy_1 \in \Gamma \subseteq \partial\Omega$. This contradiction shows that $\Gamma_h \cap \bar{\Omega} = \emptyset$. Hence Γ_h is a compact subset of $R_n - \bar{\Omega}$ if $0 < h < h_0$. This verifies (10.16).

Next we show that $T_{hy_1} F_1^* \in H_m(R_n - \Gamma_h)$. Lemma 8.2(a) yields

$$[D^p(T_{hy_1} F_1^*)] \varphi = [T_{hy_1}(D^p F_1^*)] \varphi = (D^p F_1^*)(T_{-hy_1} \varphi). \quad (10.17)$$

Note that $T_{-hy_1} \varphi \in C_0^\infty(R_n - \Gamma)$ if $\varphi \in C_0^\infty(R_n - \Gamma_h)$ and that $\|T_{-hy_1} \varphi\| = \|\varphi\|$. Hence (10.17) implies that for $|p| \leq m$ and $\varphi \in C_0^\infty(R_n - \Gamma_h)$

$$|[D^p(T_{hy_1} F_1^*)] \varphi| \leq \|F_1^*\|_{m, R_n - \Gamma} \|\varphi\|.$$

Thus we obtain $D^p(T_{h\nu_1}F_1^*) \in L_2(R_n - \Gamma_h)$ for $|p| \leq m$ and hence $T_{h\nu_1}F_1^* \in H_m(R_n - \Gamma_h)$. Since

$$(\tau_{h\nu_1}F_1)\varphi = (T_{h\nu_1}F_1^*)\varphi \quad \text{for} \quad \varphi \in C_0^\infty(\Omega_0) \quad (10.18)$$

and $\Omega_0 \subset R_n - \Gamma_h$ if $0 < h < h_0$, we have $\tau_{h\nu_1}F_1 \in H_m(\Omega_0)$ for $0 < h < h_0$. We show that

$$\|\tau_{h\nu_1}F_1 - F_1\|_{m,\Omega_0} \rightarrow 0 \quad \text{as} \quad h \rightarrow 0. \quad (10.19)$$

In fact, assume that $|p| \leq m$. Since $F_1^* \in H_m(R_n - \Gamma)$, there exists for any given $\epsilon > 0$ a $\psi \in C_0^\infty(R_n - \Gamma)$ such that

$$\|D^pF_1^* - \psi\|_{L_2(R_n - \Gamma)} < \epsilon. \quad (10.20)$$

By (10.18) we have for $\varphi \in C_0^\infty(\Omega_0)$

$$\begin{aligned} [D^p(\tau_{h\nu_1}F_1)]\varphi &= (-1)^{|p|} (T_{h\nu_1}F_1^*) (D^p\varphi) = (-1)^{|p|} F_1^* [T_{-h\nu_1}(D^p\varphi)] \\ &= (-1)^{|p|} F_1^* [D^p(T_{-h\nu_1}\varphi)] = (D^pF_1^*) (T_{-h\nu_1}\varphi) \end{aligned}$$

and hence for $0 < h < h_0$

$$\begin{aligned} \left| [D^p(\tau_{h\nu_1}F_1)]\varphi - \int \varphi T_{h\nu_1}\psi \, dx \right| &= \left| (D^pF_1^*) (T_{-h\nu_1}\varphi) - \int \psi T_{-h\nu_1}\varphi \, dx \right| \\ &\leq \|D^pF_1^* - \psi\|_{L_2(R_n - \Gamma)} \cdot \|T_{-h\nu_1}\varphi\| < \epsilon \|\varphi\|. \end{aligned}$$

Here we have used that $\text{supp}(T_{-h\nu_1}\varphi) \subset R_n - \Gamma$ since $\varphi \in C_0^\infty(\Omega_0)$ and $0 < h < h_0$. This shows that $\|D^p(\tau_{h\nu_1}F_1) - T_{h\nu_1}\psi\|_{L_2(\Omega_0)} < \epsilon$ for $0 < h < h_0$ and hence, by (10.20), (2.23), and the triangle inequality,

$$\|D^p(\tau_{h\nu_1}F_1) - D^pF_1\|_{L_2(\Omega_0)} < 2\epsilon + \|T_{h\nu_1}\psi - \psi\| < 3\epsilon$$

for sufficiently small h . This verifies (10.19).

By (10.19), we can choose for any given $\epsilon > 0$ an h with $0 < h < h_0$ such that

$$\|F_1 - \tau_{h\nu_1}F_1\|_{m,\Omega_0} < \frac{\epsilon}{2}. \quad (10.21)$$

Since $\text{dist}(\Gamma_h, \Omega) > 0$ by (10.16), there exists an $\eta \in C_0^\infty(R_n - \Gamma_h)$ such that $\eta = 1$ in Ω_0 . Assume that $\varphi \in C_0^\infty(\Omega_0)$. Then we have $\eta \cdot \varphi = \varphi$ and hence, by (10.18),

$$(\tau_{h\nu_1}F_1)\varphi = (T_{h\nu_1}F_1^*)\varphi = (T_{h\nu_1}F_1^*)(\eta \cdot \varphi) = (\eta \cdot T_{h\nu_1}F_1^*)\varphi.$$

Thus we obtain $\tau_{hy_1} F_1 = \eta \cdot T_{hy_1} F_1^*$ in $L_2(\Omega_0)$. Since $\eta \in C_0^\infty(R_n - \Gamma_h)$ and $T_{hy_1} F_1^* \in H_m(R_n - \Gamma_h)$, Theorem 9.2 yields $\eta \cdot T_{hy_1} F_1^* \in \dot{H}_m(R_n - \Gamma_h)$. Hence there exists a $\varphi_\epsilon \in C_0^\infty(R_n - \Gamma_h)$ such that

$$\|\eta T_{hy_1} F_1^* - \varphi_\epsilon\|_{m, R_n - \Gamma_h} < \frac{\epsilon}{2}.$$

Lemma 4.3 implies that

$$\|\tau_{hy_1} F_1 - \varphi_\epsilon\|_{m, \Omega_0} = \|\eta T_{hy_1} F_1^* - \varphi_\epsilon\|_{m, \Omega_0} < \frac{\epsilon}{2}$$

and hence, by (10.21), $\|F_1 - \varphi_\epsilon\|_m < \epsilon$ in $H_m(\Omega_0)$. The sequence $f_k^1 = \varphi_{1/k}$ provides the required approximation for $F_1 = \xi_1 F$ in $H_m(\Omega_0)$. The remaining components $F_2, \dots, F_N$ of F can be approximated by the same method. This completes the proof of Theorem 10.2.

11. IMBEDDING THEOREMS

In this section we shall prove several variants of Sobolev's imbedding theorem (for the original formulation see [7] and [8]) by using the concepts developed in this paper. The first two theorems hold for arbitrary open sets $\Omega \subseteq R_n$:

THEOREM 11.1. *Assume that $F \in \dot{H}_{m+j}(\Omega)$ where $m > n/2$ and $j \geq 0$. Then there exists a function $u \in C^j(\bar{\Omega})$ such that*

$$F\varphi = \int u\varphi \, dx \quad \text{for every} \quad \varphi \in C_0^\infty(\Omega). \quad (11.1)$$

THEOREM 11.2. *Assume that $F \in H_{m+j}(\Omega')$ for every $\Omega' \subset \subset \Omega^3$ where $m > n/2$ and $j \geq 0$. Then there exists a function $u \in C^j(\bar{\Omega})$ such that (11.1) holds.*

Lemma 4.3 shows that the assumption of Theorem 11.2 is satisfied if $F \in H_{m+j}(\Omega)$. (11.1) says that F is generated by u . Hence Theorems 11.1 and 11.2 imply that the linear spaces $\dot{H}_{m+j}(\Omega)$ and $H_{m+j}(\Omega)$ can be imbedded into $C^j(\bar{\Omega})$ and $C^j(\Omega)$, respectively, if u and $F = F_u$ are identified in the usual way. In order to ensure that $H_{m+j}(\Omega)$ can be imbedded into $C^j(\bar{\Omega})$, it seems to be necessary to impose some restrictions on Ω . Our third theorem contains a result in this direction. The following notation will be convenient: Assume that $y \in R_n$ and that $0 < \gamma < 1$. Then we set

$$C(y, \gamma) = \{z \in R_n : 0 < |z| < |y|, z \cdot y < \gamma |z| \cdot |y|\}, \quad (11.2)$$

$$\Gamma(y, \gamma) = \{z \in R_n : |z| = 1, z \cdot y < \gamma |z| \cdot |y|\}. \quad (11.3)$$

³ See section 0.

$C(y, \gamma)$ and $I(y, \gamma)$ are the intersections of the open cone

$$\{z : 0 < |z| < \infty, z \cdot y < \gamma |z| \cdot |y|\}$$

with the open ball $|z| < |y|$ and the unit sphere $|z| = 1$, respectively.

DEFINITION 11.1. An open set $\Omega \subseteq R_n$ has the restricted cone property if for every $x_0 \in \partial\Omega$ there exist an (open) neighborhood $O(x_0)$ of x_0 , a vector $y(x_0) \in R_n$ with $y(x_0) \neq 0$, and a number $\gamma(x_0)$ with $0 < \gamma(x_0) < 1$ such that $x + z \in \Omega$ if $x \in \bar{\Omega} \cap O(x_0)$ and $z \in C(y(x_0), \gamma(x_0))$ where $C(y, \gamma)$ is defined by (11.2).

After these preparations we formulate the last theorem of this section:

THEOREM 11.3. Assume that Ω satisfies the restricted cone property and that $F \in H_{m+j}(\Omega')$ for every bounded open subset Ω' of Ω where $m > n/2$ and $j \geq 0$. Then there exists a function $u \in C^j(\bar{\Omega})$ such that (11.1) holds.

Note that, by Lemma 4.3, the second assumption of Theorem 11.3 is satisfied if $F \in H_{m+j}(\Omega)$. In contrast to most versions of Sobolev's imbedding theorem which are available in the literature, Theorem 11.3 contains also results for unbounded domains Ω . If Ω is bounded then the restricted cone property is equivalent to the ordinary cone property (compare [1], Definition 2.1). We shall first prove Theorem 11.3 and then obtain Theorems 11.1 and 11.2 as corollaries of Theorem 11.3. The proof of Theorem 11.3 is based on the following elementary estimate:

LEMMA 11.1. Let m, n be positive integers with $m > n/2$ and R, γ be real numbers with $R > 0$ and $0 < \gamma < 1$. Then there exists a real number $c > 0$, depending only on m, n, γ , and R , such that

$$|\varphi(x_0)| \leq c \left[\int_{x_0 + C(y, \gamma)} \sum_{|p| \leq m} |D^p \varphi|^2 dx \right]^{1/2} \quad (11.4)$$

for every $\varphi \in C_0^\infty(R_n)$ and for all vectors $x_0, y \in R_n$ with $|y| = R$.

PROOF. Choose a $\zeta \in C_0^\infty(R_n)$ such that $\zeta(x) = 1$ if $|x| < R/2$ and $\zeta(x) = 0$ if $|x| > R$. Then we obtain for $x_0 \in R_n$, $\varphi \in C_0^\infty(R_n)$, and $|z| = 1$

$$\varphi(x_0) = - \int_0^R \frac{\partial}{\partial \rho} [\zeta(\rho z) \varphi(x_0 + \rho z)] d\rho$$

and hence, by integrating by parts,

$$\varphi(x_0) = \frac{(-1)^m}{(m-1)!} \int_0^R \rho^{m-1} \left(\frac{\partial}{\partial \rho} \right)^m [\zeta(\rho z) \varphi(x_0 + \rho z)] d\rho. \quad (11.5)$$

Now consider a $y \in R_n$ with $|y| = R$ and integrate (11.5) over the set $\Gamma(y, \gamma)$ defined by (11.3) with respect to z . By substituting $w = \rho z$, we get

$$\begin{aligned} \varphi(x_0) \|\Gamma(y, \gamma)\| &= \frac{(-1)^m}{(m-1)!} \int_0^R \rho^{m-1} \left[\int_{z \in \Gamma(y, \gamma)} \left(\frac{\partial}{\partial \rho} \right)^m [\zeta(\rho z) \varphi(x_0 + \rho z)] dS_z \right] d\rho \\ &= \frac{(-1)^m}{(m-1)!} \int_0^R \rho^{m-n} \left[\int_{w \in \rho \cdot \Gamma(y, \gamma)} \left(\frac{\partial}{\partial r_w} \right)^m [\zeta(w) \varphi(x_0 + w)] dS_w \right] d\rho \end{aligned}$$

where $\|\Gamma(y, \gamma)\|$ is the area of the hypersurface $\Gamma(y, \gamma)$ and $\partial/\partial r_w$ denotes the radial derivative with respect to w . Note that

$$\begin{aligned} c_1 &= \frac{1}{(m-1)!} \frac{1}{\|\Gamma(y, \gamma)\|} \left[\int_0^R \left(\int_{w \in \rho \Gamma(y, \gamma)} \rho^{2(m-n)} dS_w \right) d\rho \right]^{1/2} \\ &= \frac{1}{(m-1)!} \frac{1}{\|\Gamma(y, \gamma)\|} \left[\int_0^R \left(\int_{z \in \Gamma(y, \gamma)} \rho^{2m-n-1} dS_z \right) d\rho \right]^{1/2} \\ &= \frac{1}{(m-1)!} \frac{1}{\|\Gamma(y, \gamma)\|^{1/2}} \left[\int_0^R \rho^{2m-n-1} d\rho \right]^{1/2} \end{aligned} \quad (11.6)$$

is finite since $m > n/2$ and that $\|\Gamma(y, \gamma)\|$, and hence c_1 , does not depend on y . Schwarz's inequality implies that

$$|\varphi(x_0)| \leq c_1 \left[\int_0^R \left(\int_{w \in \rho \cdot \Gamma(y, \gamma)} \left| \left(\frac{\partial}{\partial r_w} \right)^m [\zeta(w) \varphi(x_0 + w)] \right|^2 dS_w \right) d\rho \right]^{1/2}.$$

By applying elementary differentiation rules, it follows that there exists a number $c_2 > 0$, depending only on m, n , and ζ (and hence only on m, n , and R), such that

$$\left| \left(\frac{\partial}{\partial r_w} \right)^m [\zeta(w) \varphi(x_0 + w)] \right| \leq c_2 \left[\sum_{|p| \leq m} |D^p \varphi(x_0 + w)|^2 \right]^{1/2} \quad (11.7)$$

for every $\varphi \in C_0^\infty(R_n)$ and for all $x_0, w \in R_n$. Set $c = c_1 c_2$. Then we obtain, by (11.2-3),

$$\begin{aligned} |\varphi(x_0)| &\leq c \left[\int_0^R \left(\int_{w \in \rho \cdot \Gamma(y, \gamma)} \sum_{|p| \leq m} |D^p \varphi(x_0 + w)|^2 dS_w \right) d\rho \right]^{1/2} \\ &= c \left[\int_{x_0 + C(y, \gamma)} \sum_{|p| \leq m} |D^p \varphi|^2 dx \right]^{1/2}. \end{aligned} \quad (11.8)$$

This completes the proof of Lemma 11.1.

As the next step in the proof of Theorem 11.3 we show:

LEMMA 11.2. *Under the assumptions of Theorem 11.3, there exists for every bounded open subset Ω_0 of Ω a function $u \in C^j(\bar{\Omega}_0)$ such that*

$$F\varphi = \int u\varphi \, dx \quad \text{for} \quad \varphi \in C_0^\infty(\Omega_0). \quad (11.9)$$

PROOF. Since $\partial\Omega \cap \partial\Omega_0$ is compact, we can choose a finite number of points $x_1, \dots, x_N \in \partial\Omega$ such that

$$\partial\Omega \cap \partial\Omega_0 \subset \bigcup_{i=1}^N O_i \quad (11.10)$$

where $O_i = O(x_i)$ is the open set associated to x_i by the restricted cone property. Since

$$K = \bar{\Omega}_0 \cap \left(R_n - \bigcup_{i=1}^N O_i \right) \quad (11.11)$$

is compact and contained in Ω , we have $\text{dist}(K, \partial\Omega) > 0$. Set

$$R = \text{Min}\{|y(x_1)|, \dots, |y(x_N)|, \text{dist}(K, \partial\Omega)\}, \quad (11.12)$$

$$\gamma = \text{Min}\{\gamma(x_1), \dots, \gamma(x_N)\} \quad (11.13)$$

where $y(x_i)$ and $\gamma(x_i)$ are associated to x_i by the restricted cone property. By Lemma 11.1, there exists a real number $c > 0$ such that (11.4) holds for every $\varphi \in C_0^\infty(R_n)$ and for all $x_0, y \in R_n$ with $|y| = R$. Set

$$\Omega' = \{x \in \Omega : \text{dist}(x, \Omega_0) < 2R\}. \quad (11.14)$$

We show that

$$|\varphi(x_0)| \leq c \|\varphi\|_{m, \Omega'} \quad (11.15)$$

for every $\varphi \in C_0^\infty(R_n)$ and every $x_0 \in \bar{\Omega}_0$. Here we identify as usual φ with the functional F_φ on $C_0^\infty(\Omega')$ generated by φ . Lemma 4.3 implies that $F_\varphi \in H_m(\Omega')$. Consider a point $x_0 \in \bar{\Omega}_0$. By (11.11) x_0 lies either in K or in one of the sets $\bar{\Omega} \cap O_i$ ($i = 1, \dots, N$). Hence (11.12-13) and the restricted cone property imply that there exists a $y \in R_n$ with $|y| = R$ such that $x_0 + C(y, \gamma) \subset \Omega$ where $C(y, \gamma)$ is defined by (11.2). Since the diameter of $C(y, \gamma)$ is less than $2R$, we have even $x_0 + C(y, \gamma) \subset \Omega'$. Set $\Omega'' = x_0 + C(y, \gamma)$. Since every $u \in C(\bar{\Omega}'')$ can be approximated by a sequence

$\{u_k\}$ in $C_0^\infty(\Omega'')$ in the sense of (2.16), the argument used in the proof of Lemma 2.5 shows that

$$\|D^p \varphi\|_{L_2(\Omega'')} = \left[\int_{\Omega''} |D^p \varphi|^2 dx \right]^{1/2}. \quad (11.16)$$

Hence (11.4) implies that $|\varphi(x_0)| \leq c \|\varphi\|_{m, \Omega''}$. On the other hand, we have $\|\varphi\|_{m, \Omega''} \leq \|\varphi\|_{m, \Omega'}$ since $\Omega'' \subset \Omega'$. This verifies (11.15).

Note that the restricted cone property implies the segment property introduced in Definition 10.1. Hence, by applying Theorem 10.2 to the bounded open subset Ω' of Ω , we can choose a sequence $\{f_k\}$ in $C_0^\infty(R_n)$ such that $\|F - f_k\|_{m+j, \Omega'} \rightarrow 0$ as $k \rightarrow \infty$. By (11.15) we have

$$|D^p f_k(x) - D^p f_l(x)| \leq c \|f_k - f_l\|_{m+j, \Omega'} \quad (11.17)$$

for $|p| \leq j$ and for every $x \in \bar{\Omega}_0$. (11.17) shows that the sequences $\{D^p f_k\}$ converge uniformly in $\bar{\Omega}_0$ if $|p| \leq j$. Hence $\{f_k\}$ converges to a function $u \in C^j(\bar{\Omega}_0)$. Furthermore, since the convergence is uniform, we have

$$F\varphi = \lim_{k \rightarrow \infty} \int f_k \varphi dx = \int u \varphi dx \quad (11.18)$$

for every $\varphi \in C_0^\infty(\Omega_0)$. This concludes the proof of Lemma 11.2.

PROOF OF THEOREM 11.3. Set

$$\Omega_k = \{x \in \Omega : |x| < k\}, \quad \Omega_0 = \emptyset. \quad (11.19)$$

Lemma 11.2 implies that there exists for $k = 1, 2, \dots$ a function $u_k \in C^j(\bar{\Omega}_k)$ such that

$$F\varphi = \int u_k \varphi dx \quad \text{for} \quad \varphi \in C_0^\infty(\Omega_k). \quad (11.20)$$

If $l < k$, then

$$\int u_k \varphi dx = \int u_l \varphi dx \quad \text{for} \quad \varphi \in C_0^\infty(\Omega_l) \quad (11.21)$$

and hence, by Lemma 2.4, $u_k(x) = u_l(x)$ for $x \in \Omega_l$. This shows that the function u defined by

$$u(x) = u_k(x) \quad \text{for} \quad x \in \Omega_k - \Omega_{k-1} \quad (k = 1, 2, \dots) \quad (11.22)$$

belongs to $C^j(\bar{\Omega})$. Now consider a $\varphi \in C_0^\infty(\Omega)$ and choose k such that $\text{supp } \varphi \subset \Omega_k$. Since $u(x) = u_k(x)$ for $x \in \Omega_k$, (11.20) implies that

$$F\varphi = \int u_k \varphi dx = \int u \varphi dx. \quad (11.23)$$

This completes the proof of Theorem 11.3.

Now we deduce Theorems 11.1 and 11.2 from Theorem 11.3:

PROOF OF THEOREM 11.1. Assume that $F \in \dot{H}_{m+j}(\Omega)$ and consider the norm-invariant extension F^* of F to $L_2(R_n)$ defined by (2.24). By Lemma 4.2 we have $F^* \in \dot{H}_{m+j}(R_n)$. Since R_n clearly satisfies the restricted cone property, Theorem 11.3 implies that F^* is generated by a function $u \in C^j(R_n)$. In particular, we have $u \in C^j(\bar{\Omega})$ and

$$F\varphi = F^*\varphi = \int u\varphi \, dx \quad \text{for every } \varphi \in C_0^\infty(\Omega). \quad (11.24)$$

PROOF OF THEOREM 11.2. Set

$$\Omega_k = \left\{ x \in \Omega : |x| < k, \text{dist}(x, \partial\Omega) > \frac{1}{k} \right\}, \quad \Omega_0 = \emptyset. \quad (11.25)$$

Since $\Omega_k \subset \subset \Omega$, we can choose a function $\zeta_k \in C_0^\infty(\Omega)$ such that $\zeta_k = 1$ in Ω_k . Assume that $F \in \dot{H}_{m+j}(\Omega)$. Then Theorem 9.2 implies that $\zeta_k F \in \dot{H}_{m+j}(\Omega)$. Hence, by Theorem 11.1, $\zeta_k F$ is generated by a function $u_k \in C^j(\bar{\Omega})$. In particular, if $\varphi \in C_0^\infty(\Omega_k)$, we have $\zeta_k \cdot \varphi = \varphi$ and hence

$$F\varphi = F(\zeta_k \varphi) = (\zeta_k F) \varphi = \int u_k \varphi \, dx \quad \text{for } \varphi \in C_0^\infty(\Omega_k). \quad (11.26)$$

The argument following (11.20) shows that $u_k(x) = u_l(x)$ if $l < k$ and $x \in \Omega_l$. Thus, the function u defined by

$$u(x) = u_k(x) \quad \text{for } x \in \Omega_k - \Omega_{k-1} \quad (k = 1, 2, \dots) \quad (11.27)$$

belongs to $C^j(\Omega)$. Furthermore, for every $\varphi \in C_0^\infty(\Omega)$ we can choose k such that $\text{supp } \varphi \subset \Omega_k$. Hence (11.26) and the relation $u(x) = u_k(x)$ for $x \in \Omega_k$ imply (11.1).

12. CHANGE OF VARIABLES

Consider two open subsets Ω and Ω' of R_n and a mapping from Ω into Ω' given by

$$x \rightarrow y = f(x) = (f_1(x), \dots, f_n(x)) \quad \text{for } x \in \Omega. \quad (12.1)$$

In the following we shall always assume that the mapping f is one-to-one and that f maps Ω onto Ω' . We denote the inverse mapping of f by f^- and the components of the vector function f^- by f_j^- :

$$y \rightarrow x = f^-(y) = (f_1^-(y), \dots, f_n^-(y)) \quad \text{for } y \in \Omega'. \quad (12.2)$$

We say that f belongs to the class $C_b^k(\Omega, \Omega')$ if f is a one-to-one mapping from Ω onto Ω' and if $f_j \in C_b^k(\Omega)$ and $f_j^- \in C_b^k(\Omega')$ where f_j and f_j^- are defined by (12.1-2).

Assume that $f \in C_b^1(\Omega, \Omega')$ and that $u \in C_0(\Omega)$. Then the function $v = S_0 u$ defined by

$$v(y) = (S_0 u)(y) = u(f^-(y)) \quad \text{for } y \in \Omega' \quad (12.3)$$

belongs to $C_0(\Omega')$. S_0 is a one-to-one linear mapping from $C_0(\Omega)$ onto $C_0(\Omega')$ and the inverse S_0^{-1} of S_0 is given by

$$u(x) = (S_0^{-1} v)(x) = v(f(x)) \quad \text{for } x \in \Omega. \quad (12.4)$$

The operator S_0 describes the transition from the variables $x = (x_1, \dots, x_n)$ to the new variables $y = (y_1, \dots, y_n)$.

In the following we extend S_0 to a linear operator S from $L_2(\Omega)$ onto $L_2(\Omega')$. We start with the remark that S_0 , as an operator from $C_0(\Omega)$ onto $C_0(\Omega')$, is bounded in the L_2 -norm. In fact, we have for $u \in C_0(\Omega)$

$$\|S_0 u\|_{L_2(\Omega')}^2 = \int |S_0 u|^2 dy = \int |u(f^-(y))|^2 dy = \int |u(x)|^2 \Delta(x) dx$$

where

$$\Delta(x) = \left| \det \left(\frac{\partial}{\partial x_i} f_j(x) \right) \right|. \quad (12.5)$$

Since $f \in C_b^1(\Omega, \Omega')$, we have $\Delta \in C_b(\Omega)$ and hence

$$\|S_0 u\|_{L_2(\Omega')} \leq \|\Delta\|_\infty^{1/2} \left[\int |u|^2 dx \right]^{1/2} = \|\Delta\|_\infty^{1/2} \cdot \|u\|_{L_2(\Omega)} \quad (12.6)$$

where $\|\Delta\|_\infty$ is defined by (5.4). (12.6) shows that S_0 is bounded. Since $C_0(\Omega)$ is dense in $L_2(\Omega)$, S_0 can be uniquely extended to a bounded linear operator S from $L_2(\Omega)$ into $L_2(\Omega')$. In fact, assume that $U \in L_2(\Omega)$ and consider a sequence $\{u_k\}$ in $C_0^\infty(\Omega)$ such that $\|U - u_k\| \rightarrow 0$ as $k \rightarrow \infty$. Then (12.6) implies that the sequence $\{S_0 u_k\}$ converges in $L_2(\Omega')$. Hence SU has to be defined by the limit relation $\|SU - S_0 u_k\| \rightarrow 0$ as $k \rightarrow \infty$. Clearly, SU does not depend on the special choice of the sequence $\{u_k\}$ and the mapping $U \rightarrow SU$ is linear. By (12.6) we have

$$\begin{aligned} \|SU\|_{L_2(\Omega')} &= \lim_{k \rightarrow \infty} \|S_0 u_k\|_{L_2(\Omega')} \leq \|\Delta\|_\infty^{1/2} \cdot \overline{\lim}_{k \rightarrow \infty} \|u_k\|_{L_2(\Omega)} \\ &= \|\Delta\|_\infty^{1/2} \cdot \|U\|_{L_2(\Omega)}. \end{aligned}$$

Hence S is bounded and satisfies the estimate

$$\|SU\|_{L_2(\Omega')} \leq \|\Delta\|_{\infty}^{1/2} \cdot \|U\|_{L_2(\Omega)}. \quad (12.7)$$

If $u \in C_0(\Omega)$ and $\{u_k\}$ is a sequence in $C_0^\infty(\Omega)$ with $\|u - u_k\| \rightarrow 0$ as $k \rightarrow \infty$, then $\|Su - S_0u_k\| \rightarrow 0$, by the definition of S , and $\|S_0u - S_0u_k\| \rightarrow 0$, by (12.6), and hence $Su = S_0u$. Thus S is an extension of S_0 .

In a similar way the inverse $R_0 = S_0^{-1}$ of S_0 can be uniquely extended to a bounded operator R from $L_2(\Omega')$ into $L_2(\Omega)$. Note that $RS_0u_k = R_0S_0u_k = u_k$ for $u_k \in C_0^\infty(\Omega)$ since $S_0u_k \in C_0(\Omega')$ and $R_0 = S_0^{-1}$. Hence the limit relation $\|U - u_k\| \rightarrow 0$ implies $\|SU - S_0u_k\| \rightarrow 0$ and

$$\|RSU - RS_0u_k\| = \|RSU - u_k\| \rightarrow 0.$$

Thus we obtain $RSU = U$ for every $U \in L_2(\Omega)$ and in a similar way $SRV = V$ for every $V \in L_2(\Omega')$. These relations show that the bounded extension R of the inverse operator $R_0 = S_0^{-1}$ is the inverse of the bounded extension S of S_0 . In particular, S is a one-to-one mapping from $L_2(\Omega)$ onto $L_2(\Omega')$.

Now we prove the relation

$$(SU, V)_{L_2(\Omega')} = (\Delta \cdot U, S^{-1}V)_{L_2(\Omega)} \quad (12.8)$$

for $U \in L_2(\Omega)$ and $V \in L_2(\Omega')$. First we verify (12.8) for $u, v \in C_0(\Omega)$. Lemma 2.5 implies that

$$\begin{aligned} (Su, v)_{L_2(\Omega')} &= \int S_0u \cdot \bar{v} \, dy = \int u(f^{-1}(y)) \overline{v(y)} \, dy \\ &= \int u(x) \overline{v(f(x))} \, \Delta(x) \, dx = (u \cdot \Delta, S_0^{-1}v)_{L_2(\Omega)} \\ &= (u \cdot \Delta, S^{-1}v)_{L_2(\Omega)}. \end{aligned}$$

Now choose sequences $\{u_k\}$ in $C_0^\infty(\Omega)$ and $\{v_k\}$ in $C_0^\infty(\Omega')$ such that

$$\|U - u_k\|_{L_2(\Omega)} \rightarrow 0 \quad \text{and} \quad \|V - v_k\|_{L_2(\Omega')} \rightarrow 0$$

as $k \rightarrow \infty$. Since S and S^{-1} are bounded, it follows from Schwarz's inequality and (5.3) that

$$\begin{aligned} (SU, V)_{L_2(\Omega')} &= \lim_{k \rightarrow \infty} (Su_k, v_k)_{L_2(\Omega')} \\ &= \lim_{k \rightarrow \infty} (\Delta \cdot u_k, S^{-1}v_k)_{L_2(\Omega)} = (\Delta \cdot U, S^{-1}V)_{L_2(\Omega)}. \end{aligned}$$

By setting $V = \tilde{\psi}$ in (12.8) and applying Lemma 2.3, we obtain

$$(SU)\psi = (\Delta \cdot U, S_0^{-1}\tilde{\psi})_{L_2(\Omega)} \quad \text{for} \quad \psi \in C_0^\infty(\Omega'). \quad (12.9)$$

In a similar way we get for every $V \in L_2(\Omega')$, by interchanging the roles of Ω and Ω' ,

$$(S^{-1}V)\varphi = (\Delta^- \cdot V, S_0\tilde{\varphi})_{L_2(\Omega')} \quad \text{for} \quad \varphi \in C_0^\infty(\Omega) \quad (12.10)$$

where

$$\Delta^-(y) = \left| \det \left(\frac{\partial}{\partial y_i} f_j^-(y) \right) \right|. \quad (12.11)$$

We collect our results in the following theorem:

THEOREM 12.1. *Assume that $f \in C_b^1(\Omega, \Omega')$. Then the mapping S_0 defined by (12.3) can be uniquely extended to a one-to-one, bounded, linear mapping from $L_2(\Omega)$ onto $L_2(\Omega')$. S and the inverse mapping S^{-1} are given by (12.9) and (12.10). Furthermore S and S^{-1} are related by (12.8).*

Our next result is the following version of the chain rule for differentiation:

LEMMA 12.1. *Assume that $f \in C_b^1(\Omega, \Omega')$ and that $U \in H_1(\Omega)$. Then $SU \in H_1(\Omega')$ and*

$$\frac{\partial}{\partial y_i}(SU) = \sum_{k=1}^n \frac{\partial f_k^-}{\partial y_i} \cdot S \left(\frac{\partial}{\partial x_k} U \right). \quad (12.12)$$

PROOF. By Theorem 10.1, there exists a sequence $\{u_j\}$ in $C^\infty(\Omega) \cap H_1(\Omega)$ such that $\|U - u_j\|_1 \rightarrow 0$ as $j \rightarrow \infty$. Since $\|SU - S_0 u_j\| \rightarrow 0$ and

$$\left\| S \left(\frac{\partial}{\partial x_k} U \right) - S_0 \left(\frac{\partial}{\partial x_k} u_j \right) \right\| \rightarrow 0 \quad \text{as} \quad j \rightarrow \infty,$$

we obtain for every $\psi \in C_0^\infty(\Omega)$, by applying Lemma 2.3, Lemma 2.5, and Lemma 5.3,

$$\begin{aligned} \left[\frac{\partial}{\partial y_i}(SU) \right] \psi &= -(SU) \frac{\partial}{\partial y_i} \psi = - \left(SU, \frac{\partial}{\partial y_i} \tilde{\psi} \right)_{L_2(\Omega')} \\ &= - \lim_{j \rightarrow \infty} \left(S_0 u_j, \frac{\partial}{\partial y_i} \tilde{\psi} \right)_{L_2(\Omega')} = - \lim_{j \rightarrow \infty} \int u_j(f^-(y)) \frac{\partial}{\partial y_i} \psi(y) dy \end{aligned}$$

$$\begin{aligned}
&= \lim_{j \rightarrow \infty} \int \psi(y) \frac{\partial}{\partial y_i} [u_j(f^-(y))] dy \\
&= \lim_{j \rightarrow \infty} \sum_{k=1}^n \int \psi(y) \frac{\partial}{\partial x_k} u_j(f^-(y)) \frac{\partial}{\partial y_i} f_k^-(y) dy \\
&= \lim_{j \rightarrow \infty} \sum_{k=1}^n \left(S \left(\frac{\partial}{\partial x_k} u_j \right), \bar{\psi} \frac{\partial}{\partial y_i} f_k^- \right)_{L_2(\Omega')} \\
&= \sum_{k=1}^n \left(S \left(\frac{\partial}{\partial x_k} U \right), \bar{\psi} \frac{\partial}{\partial y_i} f_k^- \right)_{L_2(\Omega')} \\
&= \sum_{k=1}^n \left(\frac{\partial f_k^-}{\partial y_i} \cdot S \left(\frac{\partial}{\partial x_k} U \right), \bar{\psi} \right)_{L_2(\Omega')} \\
&= \left[\sum_{k=1}^n \frac{\partial f_k^-}{\partial y_i} \cdot S \left(\frac{\partial}{\partial x_k} U \right) \right] \psi.
\end{aligned}$$

This verifies (12.12). Lemma 5.1 and (12.12) imply that $(\partial/\partial y_i)SU \in L_2(\Omega')$ and hence $SU \in H_1(\Omega')$. This completes the proof of Lemma 12.1. Next we prove:

THEOREM 12.2. *Assume that $f \in C_b^m(\Omega, \Omega')$ and that $U \in H_m(\Omega)$. Then $SU \in H_m(\Omega')$. Furthermore, there exists a $\gamma_m > 0$ such that*

$$\|SU\|_{m, \Omega'} \leq \gamma_m \|U\|_{m, \Omega} \quad \text{for every } U \in H_m(\Omega). \quad (12.13)$$

COROLLARY. *If, in addition, $U \in \dot{H}_m(\Omega)$ then $SU \in \dot{H}_m(\Omega')$.*

PROOF. We prove Theorem 12.2 by induction. The case $m = 1$ is contained in Lemma 12.1. The estimate (12.13) when $m = 1$ follows from (12.12), (5.3), and the boundedness of S . Now assume that Theorem 12.2 is valid for $m \leq j$ and that $f \in C_b^{j+1}(\Omega, \Omega')$ and $U \in H_{j+1}(\Omega)$. Then (12.12) yields for every differential operator D^p of order $|p| = j$

$$\frac{\partial}{\partial y_i} D^p(SU) = D^p \left(\frac{\partial}{\partial y_i} SU \right) = \sum_{k=1}^n D^p \left[\frac{\partial f_k^-}{\partial y_i} S \left(\frac{\partial}{\partial x_k} U \right) \right]. \quad (12.14)$$

Since $(\partial/\partial x_k)U \in H_j(\Omega)$, we have, by the induction hypothesis, $S((\partial/\partial x_k)U) \in H_j(\Omega')$. Furthermore, $f \in C_b^{j+1}(\Omega, \Omega')$ implies that $(\partial/\partial y_i)f_k^- \in C_b^j(\Omega')$. Hence (12.14) and Theorem 5.1 show that

$(\partial/\partial y_i) D^p(SU) \in L_2(\Omega')$ if $|p| = j$. This proves that $SU \in H_{j+1}(\Omega')$. (12.14), (5.15), (5.3), and the estimates (12.13) for $m \leq j$ imply that (12.13) holds also for $m = j + 1$. Thus, Theorem 12.2 is valid also for $m = j + 1$, and hence for every m .

In order to prove the Corollary, choose a sequence $\{u_k\}$ in $C_0^\infty(\Omega)$ such that $\|U - u_k\|_{m,\Omega} \rightarrow 0$ as $k \rightarrow \infty$. (12.13) implies that $\|SU - Su_k\|_{m,\Omega'} \rightarrow 0$ as $k \rightarrow \infty$. Since $Su_k = S_0 u_k \in C_0^m(\Omega') \subset \dot{H}_m(\Omega')$, by Lemma 4.1, and since $\dot{H}_m(\Omega')$ is complete, we conclude that $SU \in \dot{H}_m(\Omega')$.

We conclude this section with the proof of two relations which are often useful:

LEMMA 12.2. *Assume that $f \in C_b^1(\Omega, \Omega')$ and $F, G \in L_2(\Omega)$. Then*

$$(F, G) = (\Delta^- \cdot SF, SG) = (SF, \Delta^- \cdot SG) \quad (12.15)$$

where Δ^- is defined by (12.11).

PROOF. By interchanging the roles of Ω and Ω' , (12.8) implies that

$$(S^{-1}U, V)_{L_2(\Omega)} = (\Delta^- U, SV)_{L_2(\Omega')} \quad (12.16)$$

for $U \in L_2(\Omega')$ and $V \in L_2(\Omega)$. The first relation in (12.15) follows from (12.16) by setting $U = SF$, $V = G$. The second relation is a consequence of Lemma 5.3.

LEMMA 12.3. *Assume that $f \in C_b^1(\Omega, \Omega')$, $u \in C_b(\Omega)$, and $F \in L_2(\Omega)$. Then*

$$S(uF) = S_0 u \cdot SF \quad (12.17)$$

where $S_0 u$ is defined by (12.3).

PROOF. Choose a sequence $\{f_k\}$ in $C_0^\infty(\Omega)$ such that $\|F - f_k\| \rightarrow 0$ as $k \rightarrow \infty$. Note that $S(uf_k) = S_0(uf_k) = S_0 u \cdot S_0 f_k = S_0 u \cdot Sf_k$. By (5.3), we have $\|uF - uf_k\| \rightarrow 0$ and hence, since S is bounded, $\|S(uF) - S(uf_k)\| \rightarrow 0$ in $L_2(\Omega')$. On the other hand, (5.3) and the boundedness of S imply that $\|S_0 u \cdot SF - S_0 u \cdot Sf_k\| \rightarrow 0$. By combining the last two limit relations, we obtain (12.17).

REFERENCES

1. S. AGMON. "Lectures on Elliptic Boundary Value Problems." Van Nostrand, New York, 1965.
2. L. HÖRMANDER. "Linear Partial Differential Operators." Springer-Verlag, Berlin, 1963.
3. P. D. LAX. Lecture notes. Theory of functions of a real variable. New York University 1959.

4. N. G. MEYERS AND J. SERRIN. $H = W$. *Proc. Nat. Acad. Sci. U.S.* **51** (1964), 1055.
5. F. RIESZ AND B. SZ. NAGY. "Functional Analysis." New York, 1955.
6. L. SCHWARTZ. "Théorie des Distributions, Tome I et II. Hermann 1950, 1951.
7. S. L. SOBOLEV. Sur un théorème d'analyse fonctionnelle. *Math. Sbornik* **45** (1938), 471.
8. S. L. SOBOLEV. "Einige Anwendungen der Funktionalanalysis auf Gleichungen der mathematischen Physik." Berlin 1964.
9. P. WERNER. Bemerkungen zur Theorie der L_p -Räume. *J. Reine Angew. Math.* (to appear).
10. K. YOSIDA. "Functional Analysis." Springer-Verlag, Berlin, 1966.